

Software Asset Management (SAM)

Process Guide

Paris

Ref: 0001394

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Introduction

This process guide will provide a detailed explanation on how the Software Asset Management process is enabled within the ServiceNow platform. It is intended that this process be followed as closely as possible regardless of the level of maturity of the customer. ServiceNow encourages simple, lean ITAM processes and that is reflected in the out of the box design.

Customers may add additional functionality to that offered however this should only be in scenarios where there is a required business outcome gained that could not be achieved using an out of the box method. Following this approach should also ease customer upgrade paths and the ability to expand their use of the platform.

Principles and Basic Concepts

The concepts described in this guide are to be used as a baseline for successful Software Asset Management (SAM). This guide does not contain a set of specific instructions for every SAM scenario; rather it seeks to define a procedural mechanism, which organizations can utilize as a starting point for refinement of their own Software Asset Management (SAM) process. In general, the guide is aligned with the 'out of the box' functionality provided in the ServiceNow Platform and the SAM Professional Module.

Process Scope

SAM is the process responsible for managing and optimizing the request, acquisition, deployment, maintenance, utilization, and disposal of software assets.

There are strong inter-process relationships between:

- Hardware Asset Management
- Procurement Management
- Application Portfolio Management
- Cost Management
- Contract Management
- Change Management
- Request / Catalog Management
- Request Fulfilment
- Content Library Curation

Process Objectives

The primary goal of the SAM process is to reduce IT costs and limit operational, financial, and legal risks related to the ownership and use of software through managing and optimizing software assets across their lifecycle (introduction through to retirement).

The objectives of the SAM Process are to:

- Reduce or eliminate unnecessary purchases by enabling software requests to be fulfilled from available entitlements
- Identify and record all purchased software entitlements and associated user/device allocations and subscriptions within the organization
- Assign correct license metrics, procurement information (including license costs) and supporting software contracts to the software entitlements
- Create and maintain accurate software models, client access records, and software discovery models to enable the establishment of entitlement compliance positions
- Proactively identify cost savings opportunities through removal and re-use of unused software (where terms and conditions allow)
- Capture purchase order information to record cost of software entitlement
- Accurately allocate available and acquired entitlements to users and device
- Ensure blacklisted or unlicensed software are identified and remediated
- Strengthen compliance posture against publisher license agreements
- Expedite audit response and avoid fines and financial penalties relating to non-compliance
- Provide accurate software asset information into other enterprise processes (e.g. Change Management, Technology Portfolio Management) to enable business and financial decision making
- Eliminate human system integration via email, spreadsheets, and tribal knowledge

Technology pre-requisites

To optimize software asset management process efficiencies and business outcomes through the use of the ServiceNow Platform, it is recommended that the following plugins are activated:

- Software Asset Management Professional Master [com.sn.samp.master]*
- Normalization Data Services [com.glide.data_services_canonicalization.config] - required to (a) download SAM Pro content in customer instances (b) normalize the core company table in the SN platform – the SAM Pro Software Publisher table has a dependency on core_company.
- SAM Professional SCCM Usage Integration - Microsoft SCCM 2012 v2 or 2016*
- File Based Discovery [com.snc.file_signature_normalization]
- SaaS License Management[com.sn_sam_saas] – contains integration framework and core functionality for SaaS License Management
- SaaS License Management Integrations [com.sn_sam_saas_integrations] – contains integration implementations for Box, Dropbox, DocuSign and Salesforce
- Orchestration - Client Software Distribution (CSD)*

Order of plugin install (in Orlando):

- com.sn.samp.master (note, using this loads all SAMP plugins in one step including all publisher packs)
- com.glide.data_services_canonicalization.config
- com.snc.procurement*
- com.sn_sam_spend (spend detection)
- com.sn_sam_saas
- com.sn_sam_saas_integrations
- com.sn_sam_spoke (SaaS License Connections)
- com.sn_samp_eng_app (for engineering applications, requires use of OpenLM, available in App Store)
- com.sn_samp_change (for License Change Projection capability)
- com.snc.samp_usage_sccm (MS-SCCM 2012 or MS-SCCM 2016 integration) *
- com.snc.orchestratation.client_sf_distribution (Note: Users who purchase SAMP (com.snc.samp) plugin are licensed for software uninstall functionality only, also called software reclamation)

* denotes plugins that require 'maintenance' access to activate.

Relationship with other processes

Process	Relationship description	Input	Output
Request management	Submits requests for standard and non-standard software	X	
Content library curation	Receives library content that contains normalized software product information, publisher part number library, software lifecycle information, discovery maps and processor definitions	X	
	Provides details of unmatched software installations, manually normalized Software Discovery Models, publisher part numbers and manually created software lifecycle information that require adding to the Central Software Library		X
Procurement management	Receives details of purchase orders issued to publishers/vendors from corporate procurement system	X	
	Provides details of procurement requests that require converting into purchase orders in a corporate financial system (e.g. SAP, Oracle)		X
Configuration management	Receives raw software installation data associated with a Server/Computer CI for aggregation into Software Discovery Models	X	
Change management	Receives details of software licenses/entitlements that are required to fulfil a Change Request	X	
	Reserves or assigns/allocates software entitlements		X
	Provides software entitlement information to enable the calculation of projected licensing costs when assessing Normal or Emergency Changes where there are changes to the CPU counts or CPU core counts		X
Request fulfilment	Assigns available or acquired allocations/entitlements prior to fulfilment of user-initiated Service Request		X
	Provides software model details of software that requires deploying to an end-user		X

Contract management	Receives new or renewed software license contract information	X	
	Provides software entitlement information for association to a Software License Contract		X
Technology portfolio management	Provides software model lifecycle information for association with business applications/capabilities		X
Hardware asset management	Receives details of software licenses associated with uninstalled assets to remove user/device entitlements allocations	X	

Principles and basic concepts

Policies

SAM policies are needed to establish the objectives, scope and principles that govern the process and should be considered with the change management policies because they are related.

SAM policies are dependent upon the organization's business drivers, required business outcomes, contractual requirements, and on compliance to internal as well as external standards. Areas typically addressed by policies are:

- The requirement to maintain accurate information for reporting on software assets to internal and external stakeholders.
- Level of automation to reduce errors and costs.

SAM Policies are owned and monitored by the Software Asset Management Process Owner. The Process Owner provides information to management to demonstrate overall process effectiveness and efficiency, compliance at an organizational level and compliance at a department and individual level.

Policy statement	Reason for policy	Benefits
A single SAM process will be utilized throughout the organization.	To create one consistent process used across all customer geographies.	Elimination of disparate SAM processes through adoption of a common, integrated enterprise-wide SAM Process.
All software models, entitlements, allocations and installations will be recorded and maintained in the IT Asset Repository.	One source of truth for all software asset information to simplify data management and access.	Allows for the easy generation of reports to support vendor software audits and non-compliant license positions.
Any external requirements (e.g. Sarbanes-Oxley) for SAM must be addressed in process, tool and role development.	To enforce tools and control of data elements to comply with the external requirements.	Compliance with external legislation and governance.
The SAM Process Owner is accountable for the entire process and has the authority to develop policies, procedures and work instructions pertaining to the process.	To provide a single point of accountability for the SAM process.	Ensures consistency in process execution, implementation of supporting tools and control of software asset data across the entire enterprise.

Access to update software asset records is controlled, and access rights are only given to those people who are authorized to control the data.	To prevent unauthorized changes to software asset records.	Ensures accurate and up-to-date software entitlement information during software reconciliation when only those designated to control the records have access.
Automation for software installation data collection (e.g. ServiceNow supported discovery tool) and verification should be implemented and appropriately configured.	To reduce labor costs and errors from manual data entry.	Improves ability to assure data reliability Reduces cost of manual labor
Process reviews are conducted by the Process Owner on a regular basis at the Process Owner's discretion. Reviews will focus on the process consistency, process maturity and repeatability and Key Performance Indicators (KPI).	To maximize process benefits and reduce costs in modifying incorrect data and addressing inconsistent process execution.	Identifies opportunities for process, tool and staff improvements Identifies cost of maintaining the information related to the value to the organization
The SAM Process Owner will report on software license compliance to internal and external stakeholders on a regular basis.	To ensure all stakeholders understand software license compliance position.	Identifies non-compliant software Enables remedial action prior to any formal publisher/vendor audit Identifies potential company savings through removal of unused software
Software License Contracts will be maintained in the contract repository and linked with Software Entitlements.	To enable effective management of software license contracts to track contract end dates and trigger contract renewal/extension processes	Reduction in non-compliant software caused by lapsed contracts

Software cannot be installed without authorization from IT Operations.	To ensure that only authorized software can be installed in a production environment	Reduces the amount of blacklisted and unauthorized software installations Minimizes security and network performance issues caused by the installation of unauthorized software
Software procurement is managed from a central budget and charged back to departments/cost centers.	To enable effective, organization-wide tracking of software costs.	Reduces cycle times associated with cost center-based procurement Accelerates the 'true up' process for purchasing rights to remediate non-compliant software
EUC software becomes a removal candidate if quarterly utilization falls below XX hours or if the software isn't used after a pre-defined date.	To enable the re-allocation of unused EUC software	Reduces costs associated with procuring unrequired software entitlements through re-allocating available entitlements

Security considerations

Data security is a crucial aspect of SAM as it affects several modules within ServiceNow. In addition to being effective, security controls should be easily maintained and scalable. The ServiceNow platform provides several methods for securing data including user access control rules, data policies, contextual security, and domain separation.

Process roles

Each role is assigned to perform specific tasks within the process. Within a specific process, there can be more than one individual associated with a specific role. Additionally, a single individual can assume more than one role within the process although typically not at the same time. Depending on the structure and maturity of a process, all roles described may not exist in every organization.

The following describes the main process roles defined for Software Asset Management:

Role	Description
SAM Process Owner	The SAM Process Owner's primary objective is to be accountable for the success or failure of the SAM process. The Process Owner is usually a senior manager with the ability and authority to ensure the process is rolled out and used by all stakeholders. Responsible for: - Establishing SAM process measures and targets. - Ensuring the SAM process is aligned with corporate strategy. - Establishing and communicating the process mission, goals, and objectives to all stakeholders. - Documenting and maintaining the process, policies and procedures. - Resolving any cross-functional issues. - Ensuring proper staffing and training for execution. - Directing the SAM roles. - Ensuring consistent execution of the process across the organization. - Monitoring, measuring, and reporting on the effectiveness of the process to senior management. - Identifies process redesign and improvement projects.
Software Asset Manager	Responsible for: - Managing the day to day activities of the process. - Managing key relationships with peer roles within the enterprise. This includes Business Application Owners, Service Catalogue Manager and Procurement representatives. - Ensuring quality of software asset records within the Asset Management Repository, including Software Discovery Models, Software Product Models and Software Entitlements. - Escalating any issues with the process to the SAM Process Owner. - Gathering and reporting on process metrics. - Identifying requirements to modify SAM processes. - Executing reconciliation by comparing entitlements with discovered software installations to determine effective license positions for each in-scope publisher. - Analyzing reconciliation reports to determine required actions to address non-compliant software installations. - Analyzing software usage reports and establishing removal rules for infrequently used software.

SAM Administrator	Responsible for: Creating and maintaining software asset information, including Software Models, Software Discovery Models and Software Entitlements defined within the scope of their area. Resolving unmatched or partially normalized Software Discovery Models. Managing key relationships with peer roles within the enterprise. Analyzing software asset data and creating reports. Assisting the Software Asset Manager with reconciliation and report generation. Participating in the remediation of non-compliant software installations. Determining if available entitlements can be allocated prior to initiating procurement activities.
Service Catalogue Manager	Responsible for: Working with Software Asset Management team to publish authorized software in Service Catalogue for end-user request and subsequent fulfilment. Providing the ability to request ad hoc software that is not represented as a catalogue item. Owning Service Catalogue items and managing related approval/fulfilment workflows. Providing software product information if a software model is not available for representation as a catalogue item.
Procurement Manager	Responsible for: Processing approved procurement requests for software licenses and entitlements. Providing PO information for software entitlements to the SAM Administrator. Issuing PO's processed in a third-party application (e.g. ERP systems such as SAP). Providing contract-related information to the SAM Administrator for representation in the asset repository (e.g. software model retirements, changes to upgrade/downgrade terms and conditions)
Contract Manager	Responsible for: Negotiating supplier contracts for software licenses and entitlements. Approving software license contract renewals and extensions.

	Providing software license contract information that impacts software entitlements.
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RACI Matrix

Roles and responsibilities are assigned to specific process activities.

Software Asset Management RACI

ID	Activities	Requester	Approver	Software Asset Manager	SAM Administrator	Procurement Manager	Contract Manager	Software Product Owner	Configuration Manager	Software Library Content Curator	Change Manager	Software Catalog Admin
SAM 1.0	Maintain Software Models and Lifecycles											
SAM 1.1	Identify Software Publisher and Product				A/R	C						
SAM 1.2	Create Custom Software Product			C	A/R		C					
SAM 1.3	Assign/Create Discovery Map			R	A/R							
SAM 1.4	Assign Downgrade and Upgrade Rights			R	A/R		C					
SAM 1.5	Define Suite Components			R	A/R							
SAM 1.6	Assign Software Model Lifecycles			R	A/R							
SAM 1.7	Create/Update Software Model			C	A/R			I				
SAM 1.8	Review/Update Software Model Attributes				A/R							
SAM 2.0	Request Software Entitlement											
SAM 2.1	Identify Software Requirements	A/R										

ID	Activities	Requester	Approver	Software Asset Manager	SAM Administrator	Procurement Manager	Contract Manager	Software Product Owner	Configuration Manager	Software Library Content Curator	Change Manager	Software Catalog Admin
SAM 2.2	Submit Request for Software	A/R										
SAM 2.3	Approve Request		A/R									
SAM 2.4	Check Available Entitlements			C	A/R		C				C	
SAM 2.5	Submit Request for Software Assessment	A/R						C				
SAM 2.6	Review Non-Standard Software Request	I		C	I			A/R				I
SAM 2.7	Publish Software Model to Catalogue	I										A/R
SAM 3.0	Acquire Software Entitlement											
SAM 3.1	Create Procurement Request			C	A/R		C					
SAM 3.2	Approve Procurement Request				I	A/R						
SAM 3.3	Create/Approve Purchase Order					A/R						
SAM 3.4	Issue Purchase Order				I	A/R						
SAM 3.5	Receive Notification of PO Fulfillment			C	A/R	I						
SAM 4.0	Maintain Software Entitlements											
SAM 4.1	Determine Software Model and License Type			C	A/R		C					
SAM 4.2	Assign Agreement Type and Duration			C	A/R		C					
SAM 4.3	Assign License Metric			C	A/R		C					
SAM 4.4	Assign Procurement Information			C	A/R	C	C					

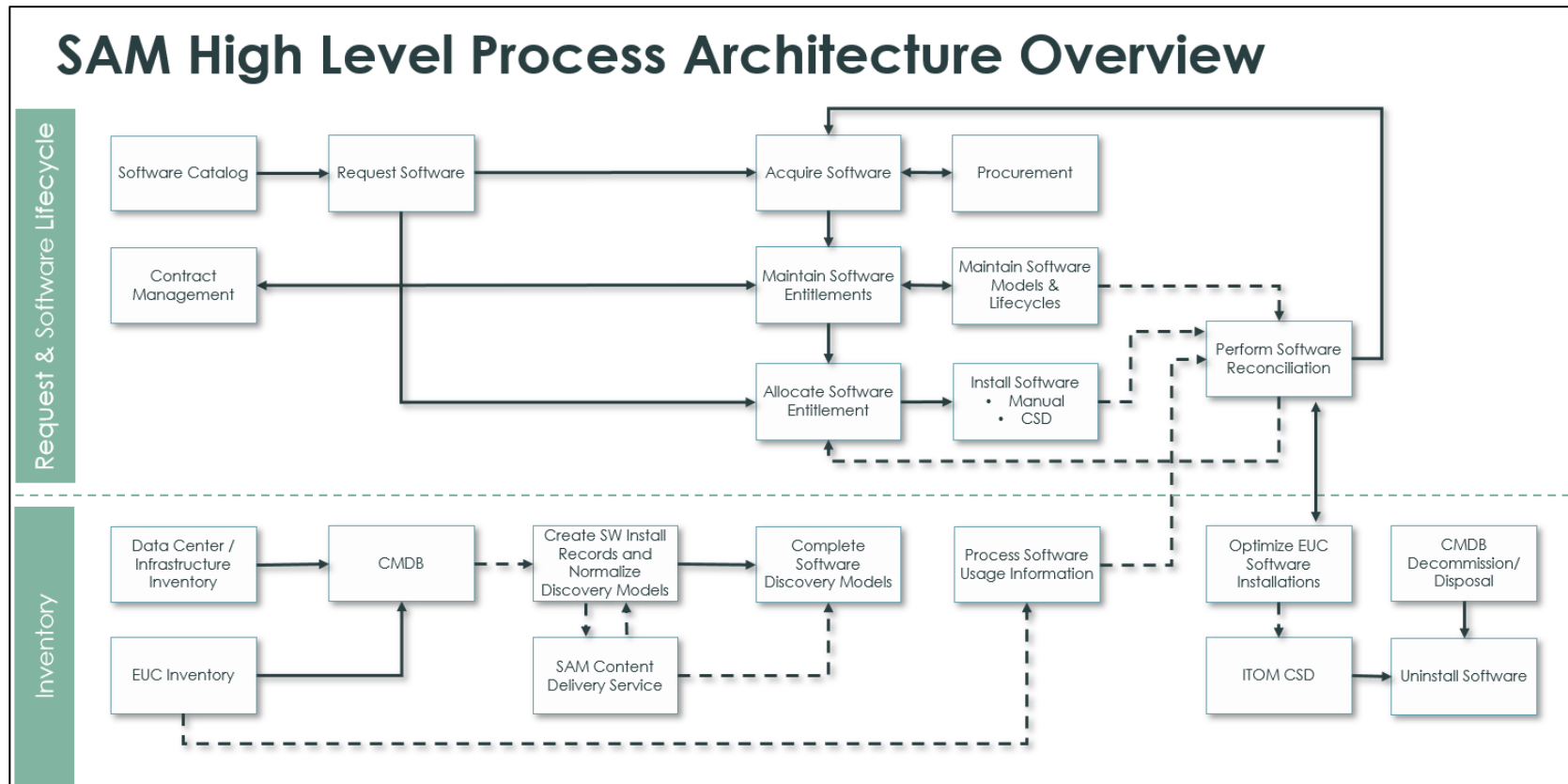
ID	Activities	Requester	Approver	Software Asset Manager	SAM Administrator	Procurement Manager	Contract Manager	Software Product Owner	Configuration Manager	Software Library Content Curator	Change Manager	Software Catalog Admin
SAM 4.5	Relate Previous Entitlement Record			C	A/R		C					
SAM 4.6	Associate Financial and Contractual Information			C	A/R							
SAM 4.7	Complete/Update Software Entitlement Record			C	A/R	C	C					
SAM 5.0	Allocate Software Entitlement											
SAM 5.1	Allocate Entitlement Rights to User/Device				A/R							
SAM 5.2	Remove Entitlement Rights from User/Device				A/R						C	
SAM 5.3	Cancel non-SaaS Software Subscription				A/R							
SAM 5.4	Reserve Entitlement Rights				A/R						C	
SAM 6.0	Create Software Installation Records and Normalize Discovery Models											
SAM 6.1	Discover CIs and Installed Software	Automated										
SAM 6.2	Create Citrix Software Installation Records	Automated										
SAM 6.3	Create/Update Software Installation Record	Automated										
SAM 6.4	Group Software Installation Records into Discovery Model	Automated										
SAM 6.5	Normalize Software Discovery Models	Automated										

ID	Activities	Requester	Approver	Software Asset Manager	SAM Administrator	Procurement Manager	Contract Manager	Software Product Owner	Configuration Manager	Software Library Content Curator	Change Manager	Software Catalog Admin
SAM 6.6	Categorize Unnormalized Software Discovery Models	Automated										
SAM 7.0	Complete Software Discovery Models											
SAM 7.1	Review Software Discovery Models			C	A/R							
SAM 7.2	Reject Normalized Software Discovery Model			C	A/R							
SAM 7.3	Resolve Partially Normalized Discovery Models			C	A/R							
SAM 7.4	Resolve Unmatched Software Installations			C	A/R							
SAM 7.5	Create Pattern Normalization Rules			C	A/R							
SAM 7.6	Review Normalization Suggestions			C	A/R							
SAM 7.7	Modify Software Discovery Model			I	A/R					I		
SAM 8.0	Perform Software Reconciliation											
SAM 8.1	Establish Software Reconciliation Scope			A/R	R							
SAM 8.2	Calculate Effective License Position	Automated										
SAM 8.3	Review Software Reconciliation Results			A/R	R							
SAM 8.4	Identify Remediation Actions for Non-Compliant Software			A/R	R							
SAM 8.5	Execute Remediation Actions	I		A/R	R	C	C	C	C			

ID	Activities	Requester	Approver	Software Asset Manager	SAM Administrator	Procurement Manager	Contract Manager	Software Product Owner	Configuration Manager	Software Library Content Curator	Change Manager	Software Catalog Admin
SAM 9.0	Optimize EUC Software Installations											
SAM 9.1	Create Reclamation Rules			A/R	R							
SAM 9.2	Produce Monthly Software Usage Report	Automated										
SAM 9.3	Create/Update Removal Candidates	Automated										
SAM 9.4	Execute Reclamation Rules & Create Software Removal Request	Automated										
SAM 9.5	Review and Approve Software Removal Request	R	A/R	I								
SAM 9.6	Validate Software Removal Request	I	I	A/R	R							
SAM 9.7	Perform Software Revocation	Automated										
SAM 10.0	Process Software Usage Information											
SAM 10.1	Analyze Usage Data			A/R	R							
SAM 10.2	Create User or Device Client Access Record for Version/Edition			A/R	R							
SAM 10.3	Update User or Device Count on Client Access Record			A/R	R							
		R: Responsible, A: Accountable, C: Consulted, I: Informed										

Software Asset Management activity description

Software Asset Management process overview



Process activity	Description
Discover Installed Hardware and Software	Discovery of raw software installation data through utilization of discovery tools such as ServiceNow Discovery or Microsoft SCCM.
Create SW Installation Records and Normalize Discovery Models	Create software installation records which are associated with the appropriate CI computer record in the CMDB. The software installation records are aggregated into Software Discovery Models that include normalized values of the discovered publisher, product and version values to enable a clear understanding of installed software for reconciliation with purchased entitlements. Non-normalized discovery models can be sent to the ServiceNow SAM Content Delivery Service (CDS) organization for normalization.
Complete Discovery Models	Reviewing Discovery Models that are less than fully normalized (if not normalized by SAM CDS). The recommended approach is to start by working with the publishers that represents the largest software spend within an organization. The focus should be on normalizing software products that are licensable. This process includes the creation of custom products and pattern normalization rules if the Software Product is not defined in SAM Pro.
Maintain Software Models and Lifecycles	Creation or modification of Software Model records that match the publisher, title and version/edition of software that requires tracking within an organization. The Software Model is mapped to a Software Discovery Model to enable reconciliation. Software Models are where software entitlement, metric attributes and discovery data is configured so that Reconciliation can compare entitlements to installations.
Request Software Entitlement	Creation and submission of requests for software. This includes requests for standard and non-standard software requests.
Acquire Software	Determination of acquisition requirements and creation / issuance of purchase orders for software entitlements.
Maintain Software Entitlements	Creation or modification of a software entitlement record that represents the number of purchased and available rights for a software product, the associated license group (e.g. Oracle, Microsoft, IBM), license metric (e.g. per named user, per device), details of user/device allocations and links to software license contracts. At a minimum, a software entitlement record must include the license type/metric, license duration and number of purchased rights.

Process activity	Description
	Software Entitlements are counted in different ways. Some are counted per device, where every installation counts as a right used. Some are counted per user, where all the installations for a user count as a right used. Sometimes the number of rights used depends on hardware characteristics, such as how many processors the device has or how many cores are in the processors. The License metric, defined on the Software Entitlement, identifies how the reconciliation process counts the rights used.
Perform Software Reconciliation	Perform reconciliation of software entitlements and discovered software to calculate the number of entitlement rights used and determine effective license positions. Only licensable Software Models will be included in Reconciliation, which will return reconciliation results. Review and analysis of reconciliation reports to determine required actions to address non-compliant software installations. This includes the creation of allocations, removal of unlicensed and unused software and purchasing new rights.
Optimize EUC Software Installations	Identification of unused or infrequently used software to establish removal and removal actions.
Allocate Software Entitlement	Just because an organization owns enough rights to cover all the installations of a software title in their environment does not mean that the software was authorized for use on all those systems. Maintaining software entitlement allocations allows for the identification of devices or users authorized (via a Request Management process) to use the software. Allocation is the first step to optimize software usage in the environment. When it is known where the software is authorized to run, installations that are not authorized can be addressed. As a leading practice, software entitlements should be allocated to a user or a device to indicate the user or device that is approved to use the software and multiple rights may be assigned to a single user or device. Some license metrics also enforce this methodology – for example, Per Named Device and Per Named User both enforce entitlements to be allocated. When these metrics are used, the software is not compliant unless entitlements have been allocated.

Software Asset Management planning and initial deployment

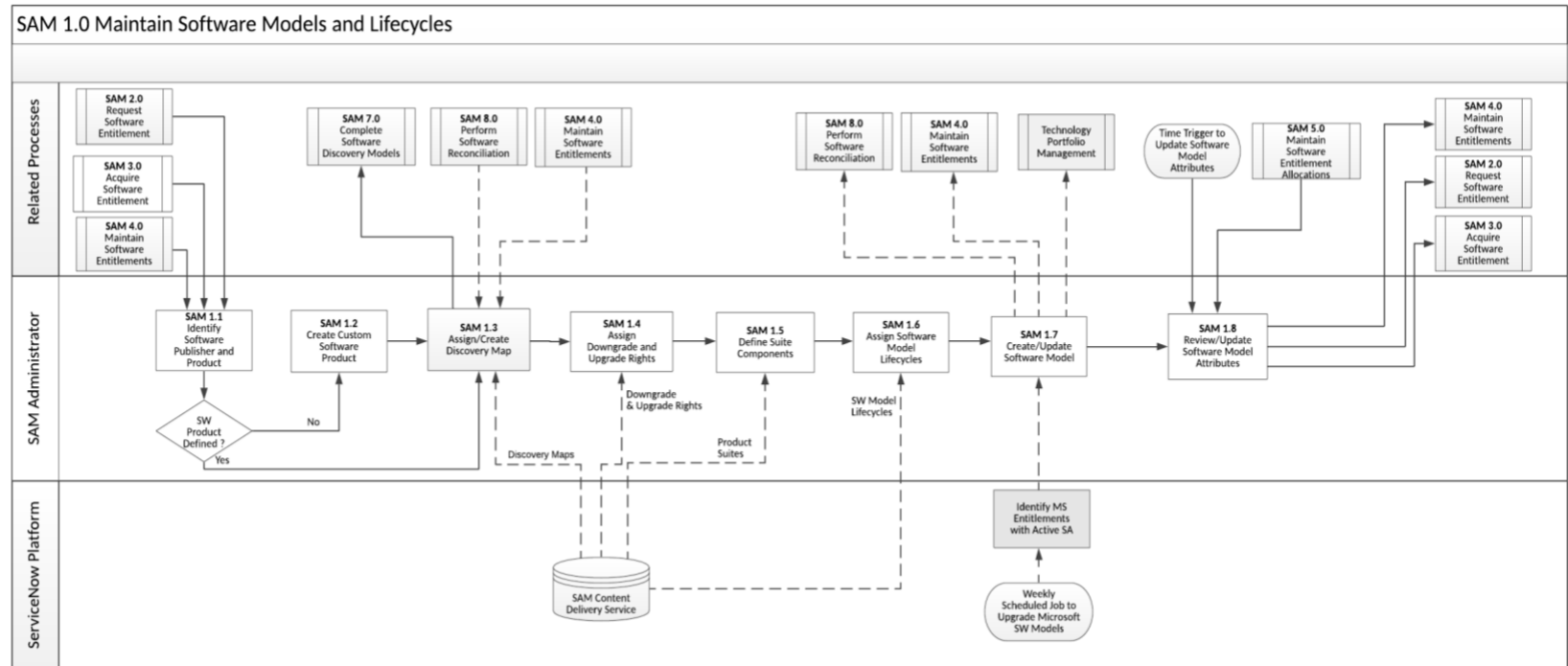
The following activities require considering prior to commencing the implementation of Software Asset Management on the ServiceNow Platform.

Activities	Description
Produce initial Software Asset Management plan	The Software Asset Management Team, under direction of SAM Process Owner, along with key stakeholders, defines the Software Asset Management Plan. The Software Asset Management Plan will be unique to each organization and is based on industry leading practices and the needs of the organization it serves. Scope Interrelated services such as Request Management, Procurement, Hardware Asset Management, Application Portfolio Management and Change Management Environments and infrastructure KPI's and Metrics Requirements Alignment to software asset management policy and business strategy Link to business, service management and contractual requirements Summarize requirements for accountability, traceability, auditability Policies Internal standards, e.g. naming conventions, change, purchasing, harvesting (re-use) Audit criteria Organization Roles and responsibilities Governance board Authorization for establishing the initial population of software asset records into ServiceNow and management of day-to-day SAM functions Processes and Procedures

Activities	Description
	See 3.1 Training and knowledge transfer Relationships and interfaces with other processes and groups Models and Entitlements Required attributes for software models and software entitlement records
Populate repository with pre-existing software installation data	Under the guidelines and structure formulated in the Software Asset Management Plan, perform an initial load of Software installation data from the organizations historic repository(s) (including migrations from the SAM Plugin) into the Now Platform.
Populate repository with pre-existing software entitlement data	One of the first steps to implement a SAM practice is to identify what you have purchased to use in your environment. This is tracked as a Software Entitlement record. There are various sources for this information, such as reseller reports and purchase orders. The use of software entitlement information in reseller reports provided by the organization's software vendors to create/populate software entitlement records is the recommended practice. This is because the software entitlement record in ServiceNow leverages a Publisher Part Number (PPN) and most organizations don't have the PPN readily in hand. The PPN is always included in the reports that the software vendor provides to the publisher under the terms and conditions of their reseller agreement. The recommended approach is to use the template to import software entitlements into ServiceNow. Refer to: https://docs.servicenow.com/bundle/paris-it-asset-management/page/product/software-asset-management2/task/view-entitlement-import-errors.html
Populate repository with pre-existing software license contract data	Under the guidelines and structure formulated in the Software Asset Management Plan, perform an initial load of Software license contract and maintenance data from the organizations historic repository(s) into the ServiceNow Platform. https://docs.servicenow.com/bundle/paris-it-service-management/page/product/contract-management/concept/c_ContractManagement.html
Populate repository with pre-existing software usage information	If the Software Asset Management Plan includes the management of usage-based licenses, including the reconciliation of Client Access Licenses, obtain user/device counts and load this information into the ServiceNow Platform.
Software Spend Detection	See attached link for details of importing financial transactions:

Activities	Description
	https://docs.servicenow.com/bundle/paris-it-asset-management/page/product/software-asset-management2/concept/software-spend-detection.html
Set up SAM Professional properties	You can set default reconciliation properties such as grouping and reconciliation debugging. Refer to: https://docs.servicenow.com/bundle/paris-it-asset-management/page/product/software-asset-management2/reference/sam-properties.html
Perform initial reconciliation	Conduct an initial software reconciliation in SAM Professional to evaluate both the Software Asset Management process and initial data integrity. Refer to: https://docs.servicenow.com/bundle/paris-it-asset-management/page/product/software-asset-management2/task/t_RunReconciliation.html
Software Asset Management Test Suite	Validate the continued functionality of Software Asset Management during application development and after upgrades. Copy and configure these automated SAM tests to identify customizations needing review. All test suites and tests should pass. https://docs.servicenow.com/bundle/paris-it-asset-management/page/product/software-asset-management2/reference/quick-tests-sam.html

Maintain software models and lifecycles



ID	Tasks	Procedures	Primary role	Input	Output
		Link from Request Software Entitlement (SAM 2.0) process because following approval of non-standard software, a software model requires creating to enable procurement and catalogue management activities.		Software model information	
		Link from Maintain Software Entitlements (SAM 4.0) process because a new Software Model is required in order to create an entitlement record (product not defined in Publisher Part Number Library).		Software model information	
		Link from Acquire Software Entitlement (SAM 3.0) process because in order to create a procurement request for a software entitlement, a software model requires creating.		Software model information	
SAM 1.1	Identify software publisher and product	When the software product is not defined in the PPN Library and no Software Model exists, initiate actions to manually create a Software Model by identifying the Publisher and Product. If the Publisher and/or Product do not exist, proceed to SAM 1.2 to create a Custom Software Product record. Otherwise, proceed to SAM 1.3.		Software model information	- Custom software product information - Software model record [WIP]
SAM 1.2	Create custom software product	Establish Publisher, Product and Product Type (e.g. licensable) and create a Custom Software Product. Identify if the product relates to subscription software. When appropriate, create a new Company record if the Publisher is not represented in the SN Platform. Note: All licensable products require a Software Model. When the custom software product is completed, proceed to SAM 1.3 and add additional information into the Software Model record.	SAM Admin	Custom software product information	- Custom software product record - Software model record [WIP]
		Link from Perform Software Reconciliation (SAM 8.0) process to create new Software Models for discovered products with unlicensed installations	Automated	Software model information	

ID	Tasks	Procedures	Primary role	Input	Output
		Link from Maintain Software Entitlements (SAM 4.0) process because a new Software Model requires creating based on the definition in the Publisher Part Number (PPN) Library.	Automated	Software model information	
SAM 1.3	Assign/create Discovery Map	Create discovery mapping conditions by associating a Discovery Map (DMAP) to the Software Model to ensure that when software reconciliation is executed, discovered software installations associated with a Discovery Model are mapped to the correct Software Model and linked entitlements. Note: this step is automated when (a) creating software entitlement records and the software model is represented as publisher part number in the SAM Content Library; (b) during software reconciliation when unlicensed installations are discovered for a software product that has no software model defined. For Software Models created using custom software products (SAM 1.2) and non-custom software products (i.e. Software Model requires creating because no Publisher Part Number is available (SAM 1.1), define the mapping conditions and add them to the Software Model in order to map the Software Model to the appropriate Discovery Model during software reconciliation. The mapping conditions include Edition/Version, Language and Platform. The assignment of the conditions (e.g. version/edition) will update the Display Name associated with the model. If a Publisher has software installation-specific licensing conditions such as whether the software is installed on a development/test/DR server OR when specific versions of a software model require excluding during license position calculations, include this information as a software install condition in the software model. Note that software install conditions can only be created for software models without a DMAP assigned.	SAM Admin/Automated	• Software model record • Discovery maps • Software model information	• Software model record [Updated] • Discovery mapping conditions [Updated]

ID	Tasks	Procedures	Primary role	Input	Output
		Link to Complete Software Discovery Models (SAM 7.0) process because following the creation of a Software Model and discovery mapping conditions associated with a custom product, pattern normalization rules require setting to enable the normalization of the Discovery Model associated with the custom product.			Software model record
SAM 1.4	Assign upgrade and downgrade rights	When a software contract allows for rights to use future or previous versions associated with the 'primary' software entitlement (i.e. no requirement to acquire upgrade rights from previous entitlements), assign the downgrade software models associated with the software model. Note: For Microsoft software models which have been assigned a DMAP via the SAM Content Service, downgrade rights are automatically populated in the software model record along with the Next Version software model (upgrade rights). When software reconciliation is executed, once the primary version is licensed, the downgrade software models will be licensed in the order they are specified on the entitlement record. Downgrade rights are used in reconciliation for the following license metrics: • Common: Per Core, Per Processor, Per User, Per Device, Per Named User or Per Named Device. • Microsoft: Per Core, Per Core with CAL.	SAM Admin/ Automated	• Software contract information • Software model record • Downgrade rights • Upgrade rights	• Software model record [Updated] • Downgrade/upgrade rights assigned
SAM 1.5	Define suite components	When applicable (i.e. product suite not created by CDS), associate suite parent Software Model or suite component Software Model(s). When appropriate, create Software Models for the parent or children Software Models if not available.		• Software model record • Product suite information	• Software model record [Updated] • Suite components assigned

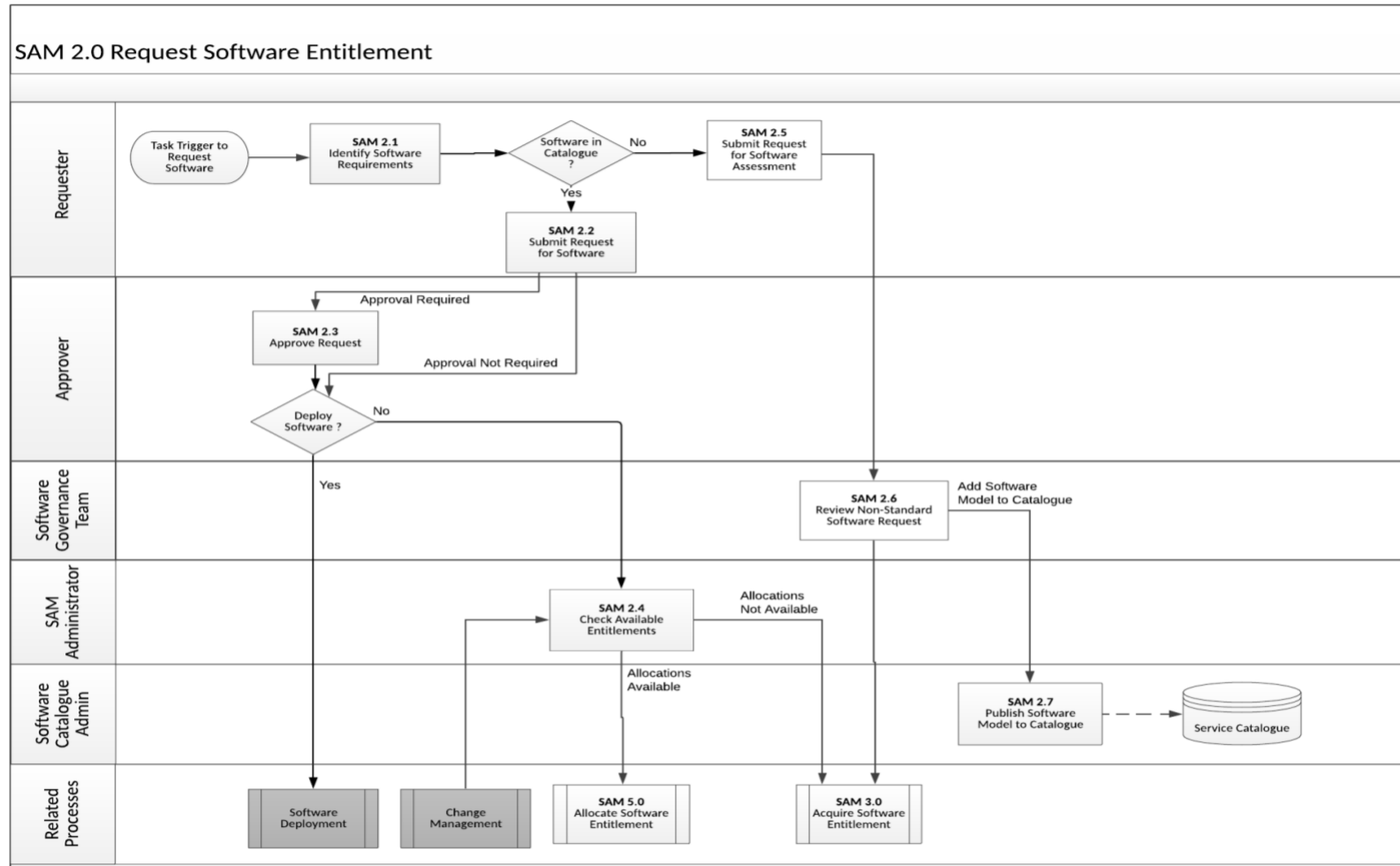
ID	Tasks	Procedures	Primary role	Input	Output
		Assign the inference percentage (if required). This is the percentage of suite components that need to be present on a system to count as a suite. Determine if a specific component in a suite must be installed to infer that the suite is installed. Choices are: Optional, Always Mandatory and Mandatory Group. For example, if the Inference percent is set to 80% and the Mandatory field is Always Mandatory for Microsoft Access, these settings specify that Microsoft Access must be installed along with three out of four other products (e.g. Word, Excel, PowerPoint, Outlook) to infer that Microsoft Office Professional is installed on a device.			
SAM 1.6	Assign software model lifecycles	If the DMAP contains a software lifecycle, this content is pulled from the content library and associated with the automatically created software model. If a software model lifecycle is not populated, determine if any lifecycle information requires linking to the software model record. Include the appropriate lifecycle types (Publisher/Internal) and phases: • Pre-Release • General Availability • Upgrade • End of Life • End of Support • End of Extended Support		• Software model record • Software model lifecycle information	• Software model record [Updated] • Software model lifecycle
SAM 1.7	Create/ update software model	Following the execution of SAM 1.1 to SAM 1.6, a software model record is created. Following creation of the software model, proceed to SAM 1.8 if additional attributes require adding to the model.	SAM Admin/ Automated	Software model record	Software model record [Completed]

ID	Tasks	Procedures	Primary role	Input	Output
		Existing Microsoft software model records are updated to a new version, following the execution of a scheduled job which identifies linked Microsoft entitlements covered by Microsoft Software Assurance (SA) which require updating to a higher version.			
		Link to Maintain Software Entitlements (SAM 4.0) process because following auto-creation of the software model record AND to cascade upgrade/downgrade rights to the software entitlement record.			Software model record
		Link to Perform Software Reconciliation (SAM 8.0) process to provide Software Model and associated data attributes into the software reconciliation process	Automated		Software model records
		Link to Technology Portfolio Management operational process because a software model lifecycle in the Software Model is available. (only applicable if the APM module is installed)			• Software lifecycle record • Software model record
		Time trigger to verify/update the software attributes of existing Software Models. This could be resultant from reports that indicate that software model attributes are incorrect.		Software model attributes require updating	
		Link from Allocate Software Entitlement (SAM 5.0) process because a software asset has been uninstalled and retired and an update is required to the software model status to reflect the software retirement.		Software retirement information	
SAM 1.8	Review/update software model attributes	For newly-created software models, establish if the model requires any additional attributes adding, including Model Status (i.e. In Production) and Business/Product Owner. Verify and determine updates to software model attributes following the creation of the Software Model OR during	SAM Admin	• Software model record • Software model	• Discovery mapping conditions • Software model updates

ID	Tasks	Procedures	Primary role	Input	Output
		the lifecycle of the Software Model to determine if any required updates. This includes: • Setting Blacklisted software indicator. • Setting License under management indicator. Reconciliation will exclude any software models which have the 'License under management' flag unchecked. • Verifying the inference percentage assigned to a software suite and the suite children associated with software suites which have been auto-created from the Content Library are correct and, when appropriate update the suite children data attributes and adjust inference percentage. • Verifying discovery mapping conditions (specifically software installation conditions) on the software model record. When applicable, perform necessary updates to the mapping conditions • Verifying the Metric Attributes to ensure the 'maximum installs' value is set correctly. When appropriate, update attribute to reflect the correct value for each license metric. • Determining if Downgrade rights require creating or updating. Full list of software model data attributes: • https://docs.servicenow.com/bundle/paris-it-service-management/page/product/product-catalog/concept/c_CreatingSoftwareModels.html		attribute validation required • Software retirement information	
		Link to Maintain Software Entitlements (SAM 4.0) process because following creation of the Software Model, additional attributes can be added to the Software Entitlement			Software model record

ID	Tasks	Procedures	Primary role	Input	Output
		Link to Request Software Entitlement (SAM 2.0) process following the completion of the required software model record.			Software model record
		Link to Acquire Software Entitlement (SAM 3.0) process following the completion of the required software model record.			Software model record

Request Software Entitlement

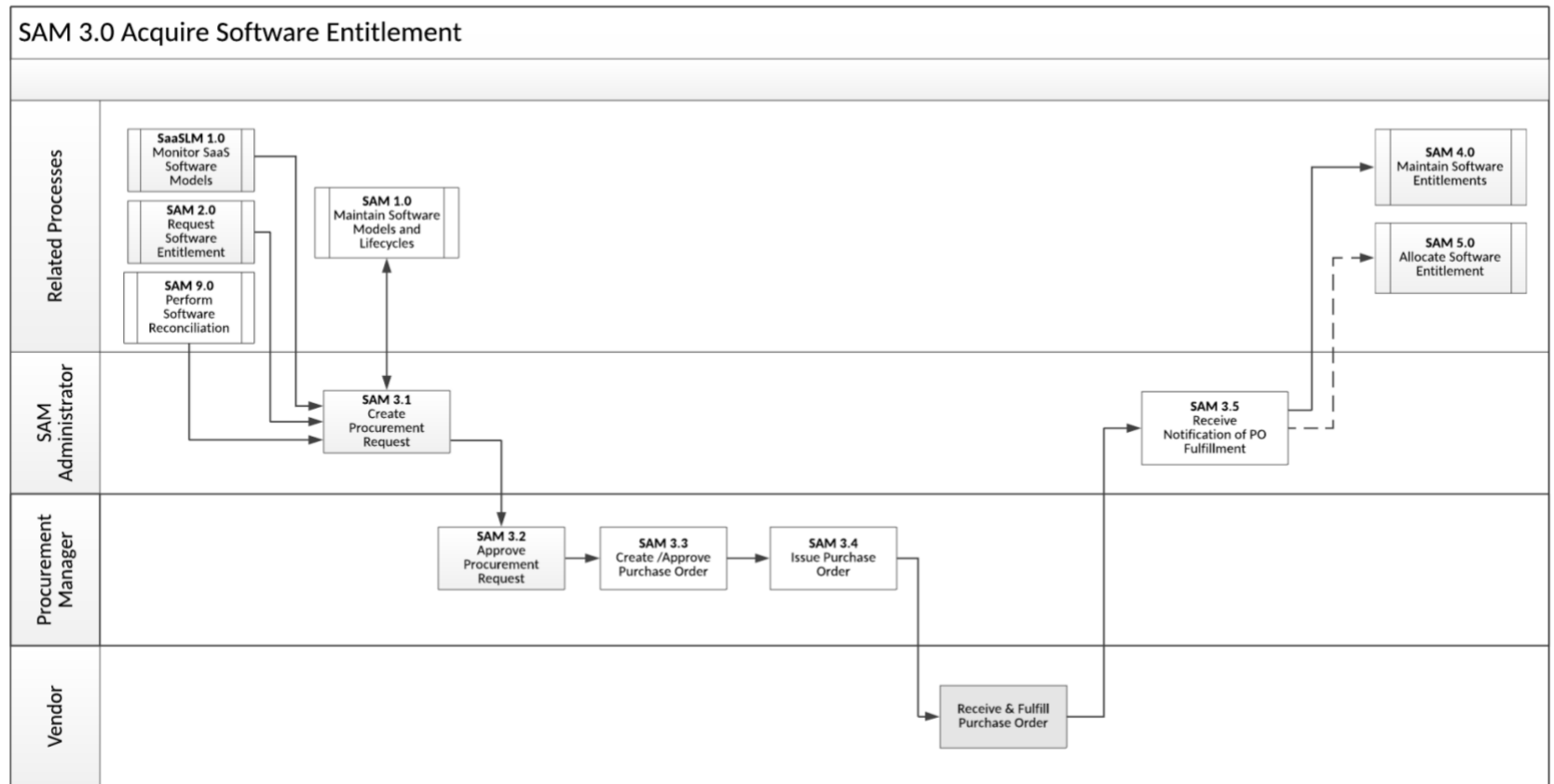


ID	Tasks	Procedures	Primary role	Input	Output
		Task trigger to submit request for software	Requester	Software entitlement requirement	
SAM 2.1	Identify software requirements	Review software catalogue (or equivalent Request Management process/system) and identify the required software model. If required software model is defined in the catalogue and available for request, proceed to SAM 2.2, otherwise proceed to SAM 2.5.	Requester	Software entitlement requirement	- Software model identified - Software model not in catalogue
SAM 2.2	Submit request for software	Create and submit the request for the software. The software model associated with the software request will determine (a) if approval is required; (b) whether a check is required to establish any entitlements are available to fulfill the request prior to any procurement activity; (c) whether the software can be deployed without an approval, entitlement check or procurement action (e.g. freeware, enterprise software)	Requester	Software model identified	- Software request requires approval - Entitlement check required - Software deployment required - Software entitlement procurement requirements
SAM 2.3	Approve request	Approve the software entitlement request. Following approval, the software model fulfillment workflow associated with the approved software request will determine (a) whether a check is required to establish any entitlements are available to fulfill the request prior to procurement and (b) whether the software can be deployed without an approval, entitlement check or procurement action.	Approver(s)	Software request requires approval	- Software entitlement request [Approved] - Entitlement check required - Software deployment required - Software entitlement

					procurement requirements
		Link to Request Fulfillment operational process because the requested software can be immediately deployed to the device/end user			Software deployment required
		Link from Change Management operational process to check for available entitlements required to fulfil a Change Request.		Entitlement check required	
SAM 2.4	Check available entitlements	Determine if the requested software entitlement is available and can be allocated to the user or device. When appropriate, login to a vendor licensing portal to determine if entitlement allocations are available. If entitlements are not available, initiate actions to acquire the entitlements. Otherwise, proceed to SAM 5.0 to allocate the entitlement.	SAM Admin	- Entitlement Check Required - Software entitlement request	- User/device allocation information - Software entitlement procurement requirements
		Link to Acquire Software Entitlement (SAM 3.0) process because there are no available license rights to allocate to the requested software entitlement.			Software entitlement procurement requirements
		Link to Allocate Software Entitlement (SAM 5.0) process to create user/device allocations for available software entitlements			User/device allocation information
SAM 2.5	Submit and monitor request for software assessment	In situations when the required software model is NOT defined in the catalogue and available for request, submit a request to the appropriate technical/software governance authority to review the non-standard software requirement and establish if the software model can be approved for use. Receive feedback from the software governance team as to whether the request has been approved or rejected.	Requester	- Software model not in catalogue - Non-standard software request reviewed	Non-standard software request

		Link from Maintain Software Models and Lifecycles (SAM 1.0) process following the creation of a software model			Software model record
SAM 2.6	Review Non-standard software request	Review non-standard software request and determine if (a) the software can be approved for use within the IT environment (b) the software model requires including in the software catalogue (c) an end-user's catalogue view requires updating so that they can request a previously approved software model. Provide feedback to the Requester following review and if the software is approved for use, initiate actions to create software models and/or publish the model into the software catalogue.	Software Governance Manager	- Non-standard software request - Software model record	- Software catalogue update required - Non-standard software request reviewed
		Link to Maintain Software Models and Lifecycles (SAM 1.0) process because following approval of non-standard software, a software model requires creating to enable procurement and catalogue management activities.		Non-standard software request reviewed	Software model information
SAM 2.7	Publish Software Model to Catalogue	Obtain information to publish the updated/completed Software Model into the Software Catalogue. This information includes: - Customer price - Software model description/image - Software fulfillment workflow - Software model supplier - Users/groups/departments/locations that can request the software	Software Catalogue	Software catalogue update required	Software Catalogue [Updated]

Acquire software entitlement



ID	Tasks	Procedures	Primary role	Input	Output
		Link from Request Software Entitlement (SAM 2.0) process because there are no available license rights to allocate to the requested software entitlement.		- Software entitlement procurement requirements - Non-standard software approved for use	
		Link from Maintain Software Models and Lifecycles (SAM 1.0) process because following the completion of the required software model record, a procurement request can be completed		Software entitlement procurement requirements	
		Link from Monitor SaaS Software Models (SaaSML 1.0) because addition SaaS license rights are required		Software entitlement procurement requirements	
		Link from Perform Software Reconciliation (SAM 9.0) process to create a purchase request to procure rights to true-up a non-compliant license position		Software entitlement procurement requirements	
SAM 3.1	Create Procurement Request	Review software requirements and determine if a quote is required from the software vendor to initiate the procurement process. If required, engage with appropriate vendor to obtain a quote. Complete the Procurement Request (ServiceNow PO record if Procurement Plugin is activated) and record the following attributes: - Vendor/Publisher - Software Model - Number of required entitlement rights - License Metric/Group - Downgrade/upgrade parameters	SAM Admin	- Software entitlement procurement requirements - Software License Contract Information - Software Vendor Quote	Procurement Request

		Maintenance/support requirements When applicable, refer to the Publisher Software License Contract to ensure the details in the Procurement Request are correct. Ensure vendor quote and any other supporting documentation is attached to the Procurement Request and submit for approval. In situations when a Procurement Request requires manually creating (e.g. to procure software required to fulfill a Change Request or Non-Standard Software Request) and the software model is not available, create software model.			
		Link to Maintain Software Models and Lifecycles (SAM 1.0) process because when creating a procurement request for a software entitlement, a software model requires creating.			
SAM 3.2	Approve procurement request	Approve the Procurement Request for the requested software and proceed to create and issue the legal purchase order.	Procurement Manager	Procurement request	Procurement request [Approved]
SAM 3.3	Create/approve purchase order	Create and approve legal PO in ERP system.	Procurement Manager	Procurement request	Purchase order [Approved]
SAM 3.4	Issue purchase order	Issue PO from ERP system to selected vendor for fulfillment.	Procurement Manager	Purchase order	Purchase order [Issued]
	Receive and fulfil purchase order	Receive PO and when appropriate provide details of information to install the software (e.g. license key).	External Vendor	Purchase order	Software entitlement information
SAM 3.5	Receive notification of PO fulfillment	Receive and validate software entitlement information from the publisher/vendor. This includes details of the PO number, license key information and number of purchased entitlement rights. Assign purchased rights to user/device (if known)	SAM Admin	Software entitlement information	Software entitlement record

		Notes: 1. if the Procurement Plugin is activated, update ServiceNow PO record and 'receive' the software entitlement. The receipt action automates the initial creation of a Software Entitlement record. 2. if the Procurement Plugin is not activated, proceed to SAM 4.1 to manually create the entitlement record.			• Allocated software entitlement • Purchase order information
		Link to Maintain Software Entitlements (SAM 4.0) process to create/update a software entitlement record.	Automated if Procurement Plugin is activated		• Software entitlement record • Purchase order information
		Link to Allocate Software Entitlement (SAM 4.0) process to create user/device allocations.	Automated if Procurement Plugin is activated		Allocated software entitlement

SAM 4.0 Maintain Software Entitlements		
SN Platform		
SAM Administrator		
Related Processes		

ID	Tasks	Procedures	Primary role	Input	Output
		Link from Maintain Software Models and Lifecycles (SAM 1.0) process because following manual or automatic creation of the Software Model, the software entitlement record can be created.		Software model record	
		Link from Create Contract Record (COM 2.0) process because a new software license contract (e.g. Enterprise License Agreement) is available for use and the software models and entitlements covered by the contract require representing in the SAM repository.		Software contract information	
		Time Trigger to create new entitlement records following the expiration of a Microsoft Software Assurance (MSA) contract.		Software contract information	
		Link from Acquire Software Entitlement (SAM 3.0) process to manually create a software entitlement record.		Purchase order information	
SAM 4.1	Determine software model and license type	When possible, use the Publisher Part Number (PPN) field to populate the Publisher, Product, Version Number/ Edition, Platform and Language of the software product. Once selected, and a corresponding software model is available, the model is automatically assigned along with the associated Discovery Map (DMAP) and other attributes associated with the DMAP – downgrade rights, product suites and software model lifecycles. If the product is available in the Publisher Part Number Library, but a corresponding Software Model record cannot be located, the software model and DMAP is created automatically. If the product is not defined in the PPN library, manually create a software model record . Following the manual creation of the software model when no PPN is available, set the License Type to Perpetual, Subscription or Upgrade. When Microsoft part is selected as a Publisher, the following license type are	SAM Admin	• Software model record • Software contract information • Publisher part number library	• Software entitlement record [WIP] • Software model information

ID	Tasks	Procedures	Primary role	Input	Output
		available: Perpetual, Software Assurance, Perpetual + Software Assurance, Subscription and Step-up are available. For upgrades from previous entitlements to new entitlements (License Type = Upgrade), select the owned entitlements that require upgrading to the new version. The owned entitlements can be for the same number or less than the number of upgrade rights purchased. Set the number of rights that need to be upgraded to the new version. This reduces the amount of active rights on the entitlement record of the software version which is being upgraded from. Note: Purchased Oracle database options require separate software entitlements. Enter details in the Database option field (visible when Product = DB Server in the software model form).			
		Link to Maintain Software Models and Lifecycles (SAM 1.0) process to manually or automatically create a new Software Model			Software model information
SAM 4.2	Assign agreement type and duration	Assign Agreement Type (e.g. ELA, Generic or publisher-specific agreement) For all Agreement Types, refer to: https://docs.servicenow.com/bundle/paris-it-asset-management/page/product/software-asset-management2/reference/agreement-types.html Assign license duration (e.g. Perpetual, Subscription). Notes: For all subscription entitlements and Microsoft entitlements with a Software Assurance, Perpetual + Software Assurance and Step-up license type, the start and end dates of the contractual period are mandatory.	SAM Admin	- Software contract information - Software entitlement record	Software entitlement record [Updated]

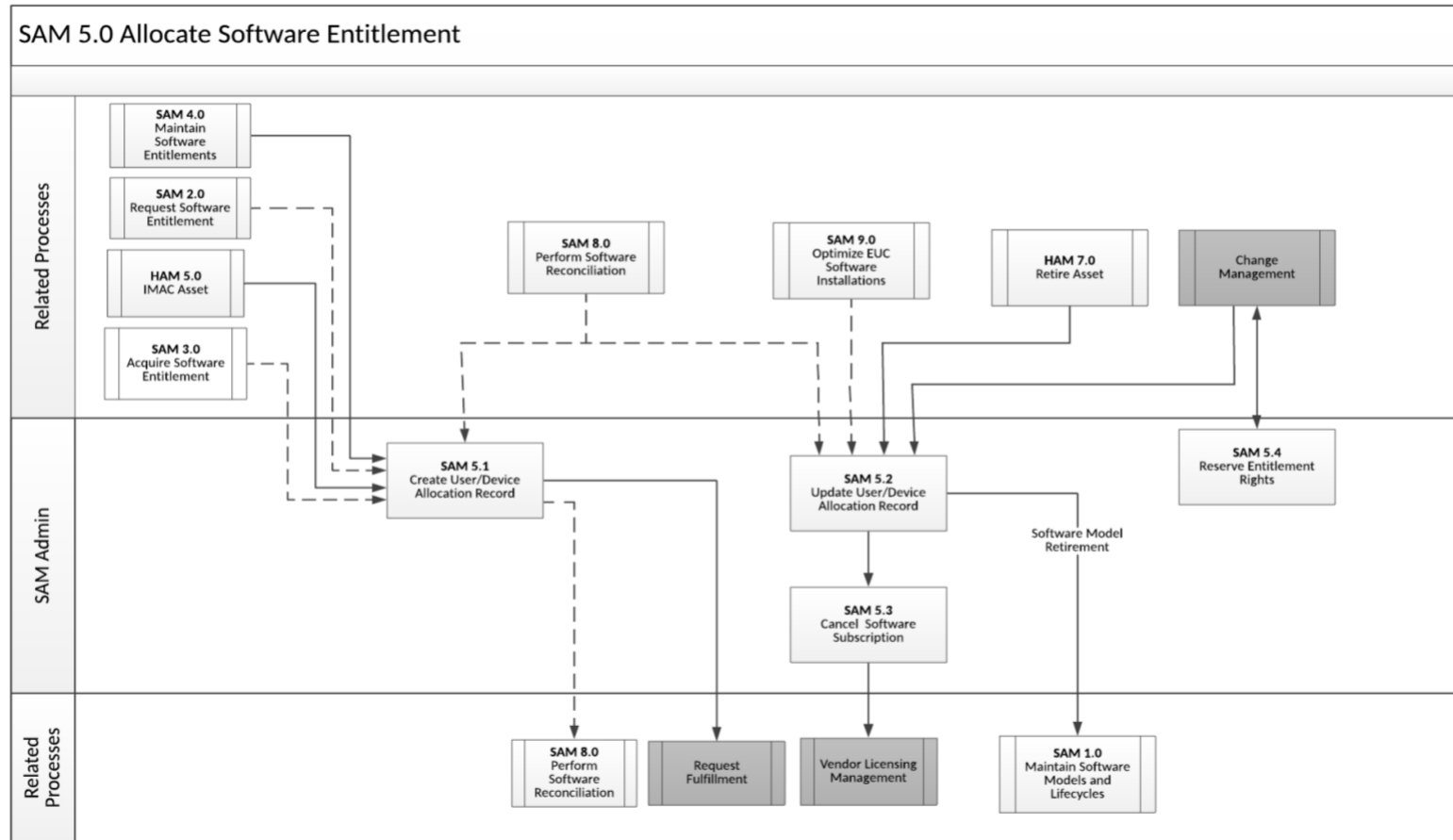
ID	Tasks	Procedures	Primary role	Input	Output
		https://docs.servicenow.com/bundle/paris-it-asset-management/page/product/software-asset-management2/task/create-entitlement-microsoft-sa.html 2. While a software entitlement for subscription software is under maintenance in the contractual period, the 'In Maintenance' box of the entitlement record is checked.			
SAM 4.3	Assign license metric	Define Metric Group (e.g. Common, Subscription, Microsoft, Adobe, Oracle, Citrix, VMWare, IBM, SAP) and License Metric (e.g. User Subscription, Per User, per Device, Per Named User, Per Named Device, User CAL, Device CAL) to calculate effective license positions during reconciliation. For all Metric Groups and License Metrics, refer to: https://docs.servicenow.com/bundle/paris-it-asset-management/page/product/software-asset-management2/reference/software-entitlement-fields.html Notes: 1. In situations when a Metric Group and/or License Metric is not defined (e.g. obscure or specialized software), a Custom License Metric can be scripted and associated with the entitlement record. 2. The selected license metric automatically associates metric attributes (e.g. how many core processors to count per license) to the software model record which is referenced during the calculation of license positions.	SAM Admin	• Software contract information • Software entitlement record • Custom license metric information	Software entitlement record [Updated]
SAM 4.4	Assign procurement information	Assign the number of rights associated with the purchased software entitlement. For Microsoft Software	SAM Admin	• Software contract information	Software entitlement record [Updated]

ID	Tasks	Procedures	Primary role	Input	Output
		Assurance renewals, enter the number of rights that require renewing. In situations when Purchased Rights are “received” following the issuance and receipt of a PO during procurement process (see SAM 3.5), this step is automated. Document unit and total cost of purchased software entitlement and other PO and invoice information (if not already pre-populated). In situations when a Microsoft SA has expired and an entitlement record has been created to reflect the reflect the new SA, proceed to SAM 4.5 to link the entitlement record associated with the expired SA.		Software entitlement record	
SAM 4.5	Link previous entitlement record	Locate the entitlement record linked to the expired software assurance agreement and enter the amount of rights that require renewing. Once the original perpetual licenses entitlement is linked, the amount of renewed rights renewed is tagged is being under an active maintenance contract. In situations when only a subset of the rights associated with the original entitlement require renewing, enter the number of rights that require renewing and this will reduce the number of rights under maintenance on the original entitlement record and create a new entitlement record (License Type = Perpetual; Maintenance flag is unchecked) for the balance of the rights not renewed.		Software entitlement record	Software entitlement record [Updated]
		Link from Acquire Software Entitlement (SAM 3.0) process to add attributes following auto-creation of the software entitlement record.		Software entitlement record	

ID	Tasks	Procedures	Primary role	Input	Output
SAM 4.6	Associate financial and contractual information	Refer to the Publisher licensing terms and conditions and establish if the entitlement rights for software compliance are for a specific company/legal entity, site/ location or department. If such conditions are stipulated, add the appropriate attributes to the Software Entitlement. Establish if any internal policies regarding the use of software by a particular department or cost center require reflecting on the Software Entitlement. If any of these dimensions are relevant, set the value accordingly. Determine if any downgrade/upgrade parameters require associating with the entitlement record (if not already pre-populated) and when applicable, add this information to the entitlement record. When appropriate, add details associated with lease contracts and warranty end-dates and link the entitlement to a software license contract		• Software entitlement record • Software contract information	• Software entitlement record [Updated] • Entitlement linked to contract
		Link to Modify Contract Record (COM 3.0) process because a new entitlement is linked to a contract.	Automated		Entitlement linked to contract
		Link from Procurement/Contract Management operational process because amendments to a publisher's licensing terms and conditions (e.g. changes to upgrade/downgrade parameters) require representing in the software entitlement records.		Publisher license amendment	
		Link from Modify Contract Record (COM 3.0) process because a software license contract has been cancelled and an update is required to the status of the entitlement records that were covered by the cancelled contract.		Software contract cancellation information	

ID	Tasks	Procedures	Primary role	Input	Output
SAM 4.7	Complete/update software entitlement record	Complete initial creation of a software entitlement record OR modify data attributes in an existing record Upon creation/update, the active rights will be updated to reflect the number of purchased rights. Other key information that requires including in the entitlement record: - Changes to location, department and cost center information - Updates to the status of the Software Entitlement (Note: Status will change to 'Retired' when the end-date associated with a subscription license passes) When applicable, proceed to SAM 5.1 to assign/allocate the entitlement to a user/device. Note: When the status of an entitlement record is updated to 'Retired' (i.e. the software license contract is cancelled, or software asset is retired/uninstalled from a production environment), the active rights will reduce to zero. Determine if the status of the Software Model requires updating.	SAM Admin	- Purchase order information - Software contract cancellation information - Software entitlement information - Publisher license amendment - Software retirement information	- Software entitlement record [Updated] - Software model updates
		Link to Allocate Software Entitlements (SAM 5.0) process to create user/device allocations.			Allocation information
		Link to Monitor SaaS Software Models (SaaSML 1.0) process to provide details of the number of rights purchased for population into the license usage summaries			Software entitlement record

Maintain software entitlement allocations



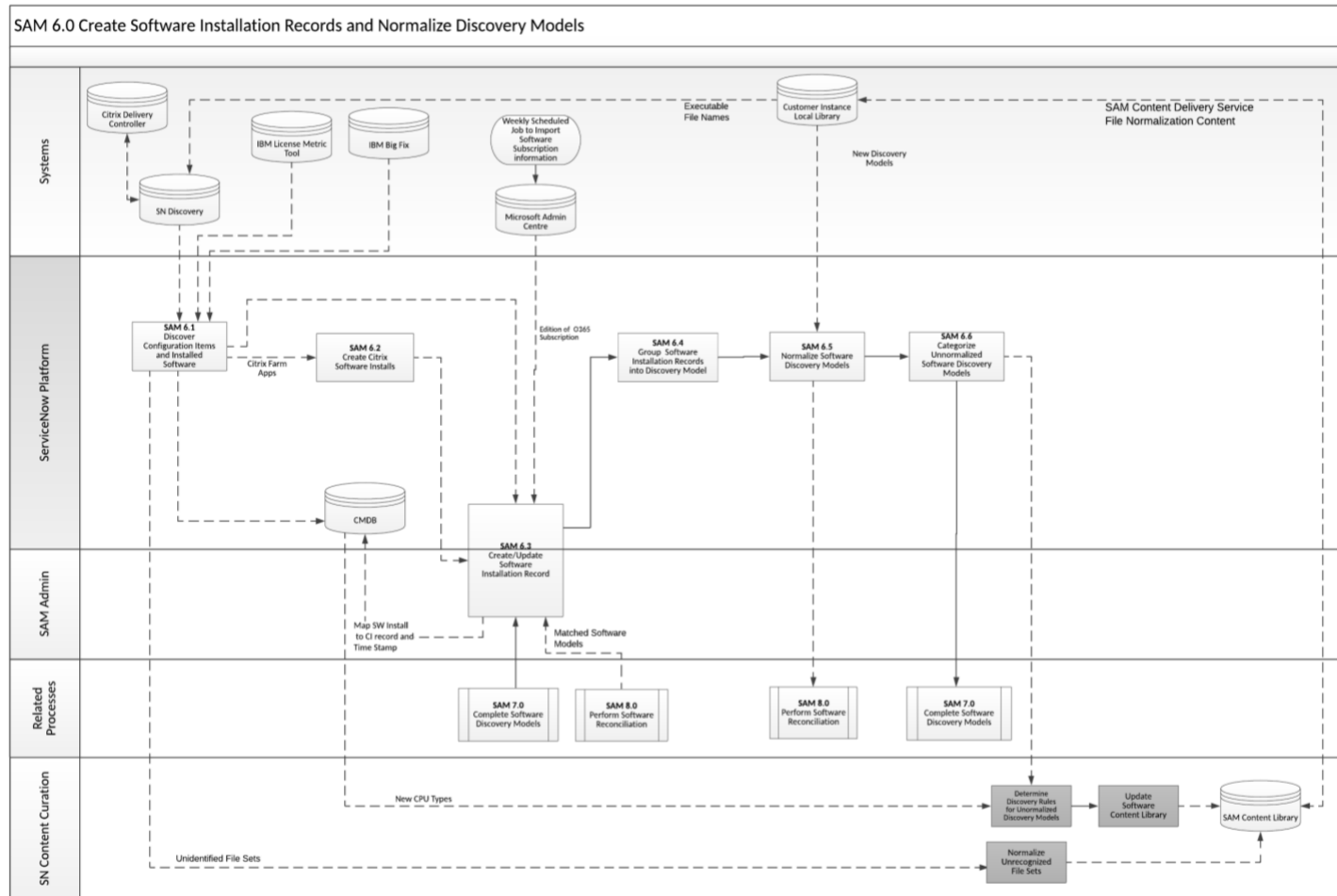
ID	Tasks	Procedures	Primary role	Input	Output
		Link from Install/Move/Add/Change Asset (HAM 5.0) process to create or remove user/device allocations following hardware swap-out		- Allocation information - Service request	
		Link from Request Software Entitlement (SAM 2.0) process to create user/device allocations for available software entitlements	Automated	- Allocation information - Service request	
		Link from Perform Software Reconciliation SAM (8.0) process to allocate entitlement rights following analysis of remediation options	Automated	Allocation information	
		Link from Maintain Software Entitlements (SAM 4.0) process to create user/device allocations		Allocation information	
		Link from Acquire Software Entitlement (SAM 3.0) process to create user/device allocations		Allocation information	
SAM 5.1	Create user/device allocation record	A user or device allocation is created from a software entitlement to specify a user or device to which rights have been allocated. Create or update user/device allocation records following the: - acquisition of entitlements that require allocating to a user/device (if Procurement Plugin is activated, user/device allocations can be added during the 'receiving' process – see SAM 3.5) - identification of available rights during Request Software process (see SAM 2.4) - execution of software reconciliation, where it was identified that a user/device allocation requires creating (automated) When allocating entitlements to a user or device, ensure the following information is updated:		- Allocation information - Service request - Change request	User/device allocation record

ID	Tasks	Procedures	Primary role	Input	Output
		- Quantity of entitlements allocated (note: multiple rights become relevant where many rights are needed to fully license a device, such as with a 'per core' license metric). - Assigned To (user) - Allocated To (device) - Assigned Date - License Key (when applicable) Refer to PO information, Service/Change Request or usage/subscription reports to ensure that the allocations are created correctly. When appropriate, engage with the user or CI Administrator to verify that the entitlements are associated with the correct user/device. Note: in situations where cloud licenses are tracked in a vendor portal (e.g. Adobe CC/Office 365), allocate license to user/device in the portal.			
		Link to Request Fulfillment operational process to fulfill request to deploy software that is allocated to a device or user.			Allocated software entitlement
		Link from Retire Asset (HAM 7.0) process to remove device/user allocations following asset retirement		- Unused software entitlement information - Service request	
		Link from Perform Software Reconciliation (SAM 8.0) process to unallocate rights following analysis of remediation options and software usage	Automated	Allocation information	
		Link from Optimize Software Installations (SAM 9.0) process to unallocate rights following software revocation	Automated	Unused software entitlement information	

ID	Tasks	Procedures	Primary role	Input	Output
		Link from Change Management process to remove device/user allocations following change implementation		Unused software entitlement information Change request	
SAM 5.2	Update user/device allocation record	A user or device allocation is removed from a software entitlement usually occurs following: • Retirement/transfer of hardware assets (including employee off-boarding) during Request Fulfillment or Change Management processes • Retirement/uninstall of software models based upon an organization's technology policy. • Software reconciliation, where it was identified that an allocation requires removing (automated). • Analysis of software usage that identified unused entitlements In situations where a software model is uninstalled due to retirement, initiate actions to update the Status value in the Software Entitlement and Software Model records. If the removed entitlement relates to a user subscription license, proceed to SAM 5.3 to deactivate the license.	SAM Admin	• Unused software entitlement information • Service request • Change request	• Software model updates • User/device allocation record [Updated] • Software retirement information
		Link to Maintain Software Models and Lifecycles (SAM 1.0) process because a software asset has been uninstalled and retired and an update is required to the status of the software model and linked entitlement records associated with the retired software.			Software retirement information
SAM 5.3	Cancel software subscription	Perform necessary actions to de-activate user software subscription licenses in the event of an employee changing roles or leaving the company. This may require logging into a vendor cloud/SaaS license management system and updating user or role subscription information.	SAM Admin	Software retirement information	Cancelled software subscription

ID	Tasks	Procedures	Primary role	Input	Output
		Link from Change Management operational process because available entitlement rights require reserving during Change Assessment so that the rights are available following Change approval and used during implementation.		- Allocation information - Change request	
SAM 5.4	Reserve entitlement rights	Review Change Request and reserve the software entitlement rights required for Change implementation. Determine how long the rights require reserving for.	SAM Admin	- Allocation information - Change request	Software entitlement record [Updated]
		Link to Change Management operational process to notify Change Manager that the required software entitlement rights are reserved.			

Create software installation records and normalize discovery models



ID	Tasks	Procedures	Primary role	Input	Output
SAM 6.1	Discover Configuration Items and Installed Software	Based on pre-determined schedules, execute discovery events to discover physical and virtual computers and associated software installations. Discovered CIs are either created or updated into the CMDB. If Discovery system identifies new CPU Types (based on a combination of Processor Type/Model and Server Model) that are not mapped to a Processor Definition defined by a Publisher, this information is pushed to Content Curation Team for mapping and inclusion in the SAM Content Delivery Service (CDS). Discovery also recognized file sets that are installed (not running and not identifiable by probes and patterns) and their properties, as an input to a SAM API in Discovery which maps/normalizes the discovered files with software packages and identifies normalized publisher, product, version and edition. File sets normalized by the SAM API automatically create software installation records (Discovery Source = SN File Discovery). files can be manually normalized or sent to the ServiceNow CDS for normalization.	Automated		- Discovered software installations - Discovered Citrix software installations - Discovered computers - Unmatched CPU types - Recognized File Sets
SAM 6.2	Create Citrix Software Installs	Discovery captures delivery group mappings to published apps and desktops in Citrix. Delivery groups are mapped to AD groups which determines the users that have potential access to the published apps. For each user and device (s) assigned to it, software installation records are created for the applications that are mapped to the delivery group. Citrix software installation records (Discovery Source = Citrix) are created via a weekly scheduled job. The software installation record is stamped with the date when the application in the Citrix farm was last used.	Automated	- Discovered computers - Discovered Citrix software installations	Citrix software installation records

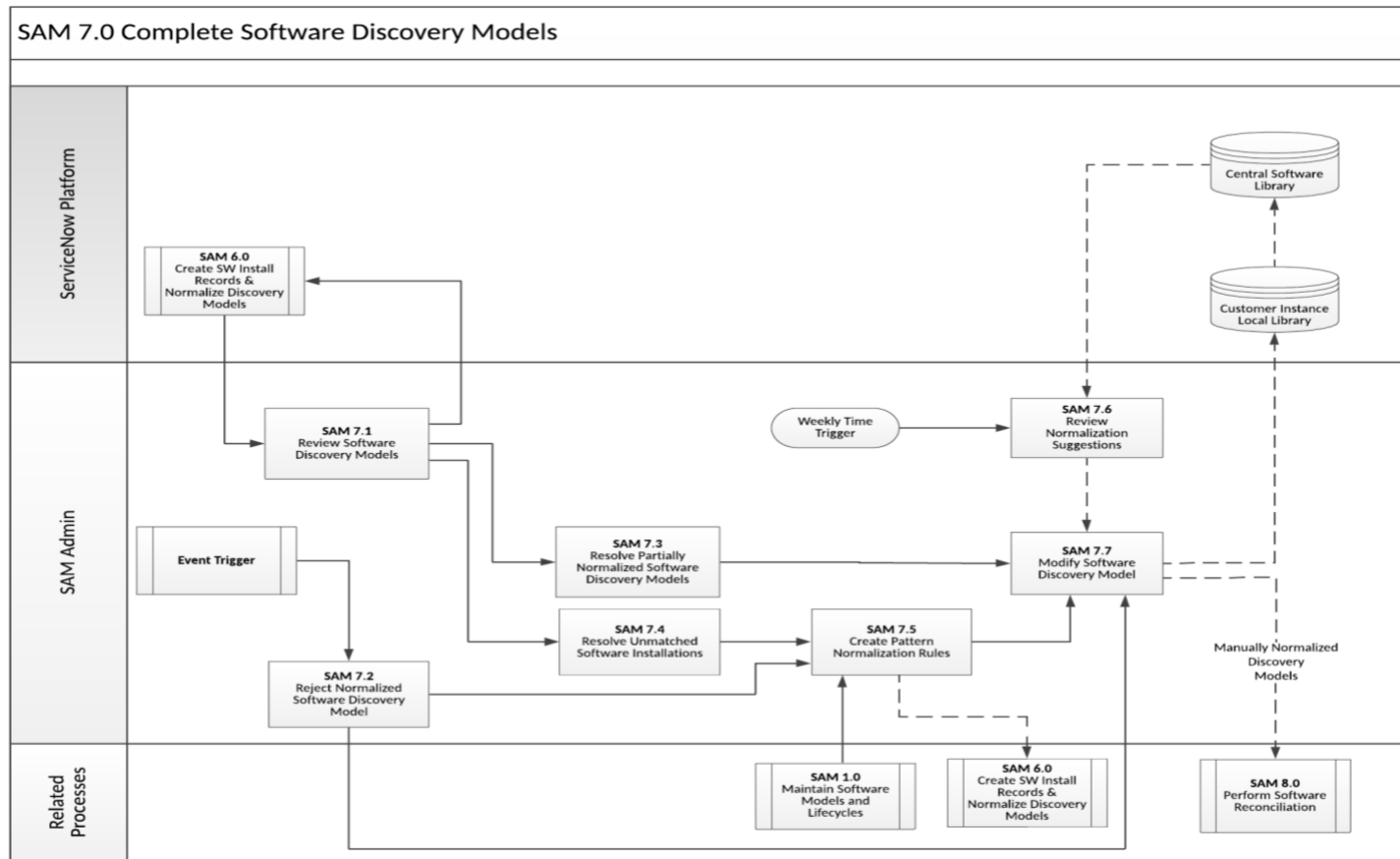
ID	Tasks	Procedures	Primary role	Input	Output
		Link from Weekly Scheduled Job to update the Edition value on Software Installation records related to Office 365 software subscriptions.	Automated	Subscription edition information	
		Link from Complete Discovery Models (SAM 7.0) process to add Edition value to an existing software installation record.		Software installation edition identified	
		Link from Perform Software Reconciliation (SAM 8.0) process because following reconciliation of software products with linked downgrade rights, a specific software model is associated with the software installation record	Automated	Software Model matched with Software Installation Record	
SAM 6.3	Create/update software installation record	Create or update software installation records that represent software installed within an organization's network environment. Software installation records are associated with the Physical or Virtual CI record in the CMDB. Software installation records are updated in the following situations: - Undiscovered Edition value requires adding to the software installation record (automated for Office 365 installation records) - During review of Discovery Models, an Edition value required adding to a software installation record	Automated/SAM Admin	- Discovered software installations - Citrix software installation records - Subscription edition information - Software installation edition identified	- Software installation record [New/Updated] - Software discovery model information
SAM 6.4	Group software installation records into discovery model	Aggregate the software installation records, created in SAM 6.3, that have the same publisher, product and version into the same Software Discovery Model (Note: Newly discovered software installations inherit the normalized values in the corresponding Discovery Model, if one exists).	Automated	Software installation records	Software discovery model [WIP]

ID	Tasks	Procedures	Primary role	Input	Output
		If a Software Discovery Model doesn't exist, one is automatically created for each software installation that has the same discovered publisher, product and version. Note: For IBM software installation data that is provided via the ILMT integration, only the Discovered Publisher and Discovered Product are required to form a Software Discovery Model. No product version will be discovered. Such Discovery Models require manually normalizing.			
		Link from Complete Software Discovery Models (SAM 7.0) process to provide pattern normalization rules for custom products into the normalization process		Pattern normalization rules	
SAM 6.5	Normalize software discovery models	Utilizing the OOTB normalization rules (updated weekly via CDS) and via manually created pattern normalization rules, the Discovery Model is updated to include the normalized product, publisher and version values. The daily normalization scheduled job also identifies the following additional information that will be populated into the Software Discovery Model: • Product Type (Licensable, Patch, Driver, Child) • Platform of Discovered Software • Language of Discovered Software • Edition of Discovered Software • Full Version Number On occasion, a software title will not normalize. In these situations, proceed to SAM 6.6.	Automated	• Software discovery model • Local software library • Pattern normalization rules	Completed normalization process
		Link to Perform Software Reconciliation (SAM 8.0) process to provide normalized Software Discovery	Automated		Normalized discovery model

ID	Tasks	Procedures	Primary role	Input	Output
		Models to match up with entitlements purchased to calculate effective license positions.			
SAM 6.6	Categorize unnormalized discovery models	If the Discovery Model cannot be normalized or fully normalized with Publisher, Product and Version values, an attempt is made to partially normalize the Discovery Model using the Discovered Publisher and Discovered Product values. If a match is found, the Publisher and Product values are set, normalization status is set to Partially Normalized and the rest of the Discovery Model (Version and additional information such as Platform and Edition) requires manual update. If the Discovery Model cannot be normalized using the Discovered Publisher and Discovered Product values, the Discovered Publisher is validated in the normalization engine. If a match is found, the Publisher value is set, normalization status is set to Publisher Normalized and the rest of the Discovery Model (Discovered Product/ Version and additional information) requires manual update. If no Publisher match is found, the normalization status is set to Match Not Found. Note: If only Publisher and Product fields are normalized, and the Product Type is 'Not Licensable', 'Child', 'Driver' or 'Patch', the normalization status is Normalized. The unnormalized or partially normalized Discovery Models can be reviewed/analyzed for remediation (SAM 7.0 – Complete Discovery Models) or if no action is taken, the Discovery Models with Partially Normalized, Publisher Normalized or Match Not Found statuses are transferred to the ServiceNow SAM Content Delivery Service so that the ServiceNow Content Curation Team can create new discovery	Automated	Completed normalization process	Unnormalized discovery model

ID	Tasks	Procedures	Primary role	Input	Output
		rules (assuming that the organization has opted in to the SAM Content Delivery Service).			
		Link to Complete Software Discovery Models (SAM 7.0) process to review partially normalized Software Discovery Models following the normalization process.			Partially normalized discovery model
		Link to Complete Software Discovery Models (SAM 7.0) process to create Software Discovery Models for software installations that could not be normalized during the automated normalization process			Unnormalized discovery model

Complete software discovery models



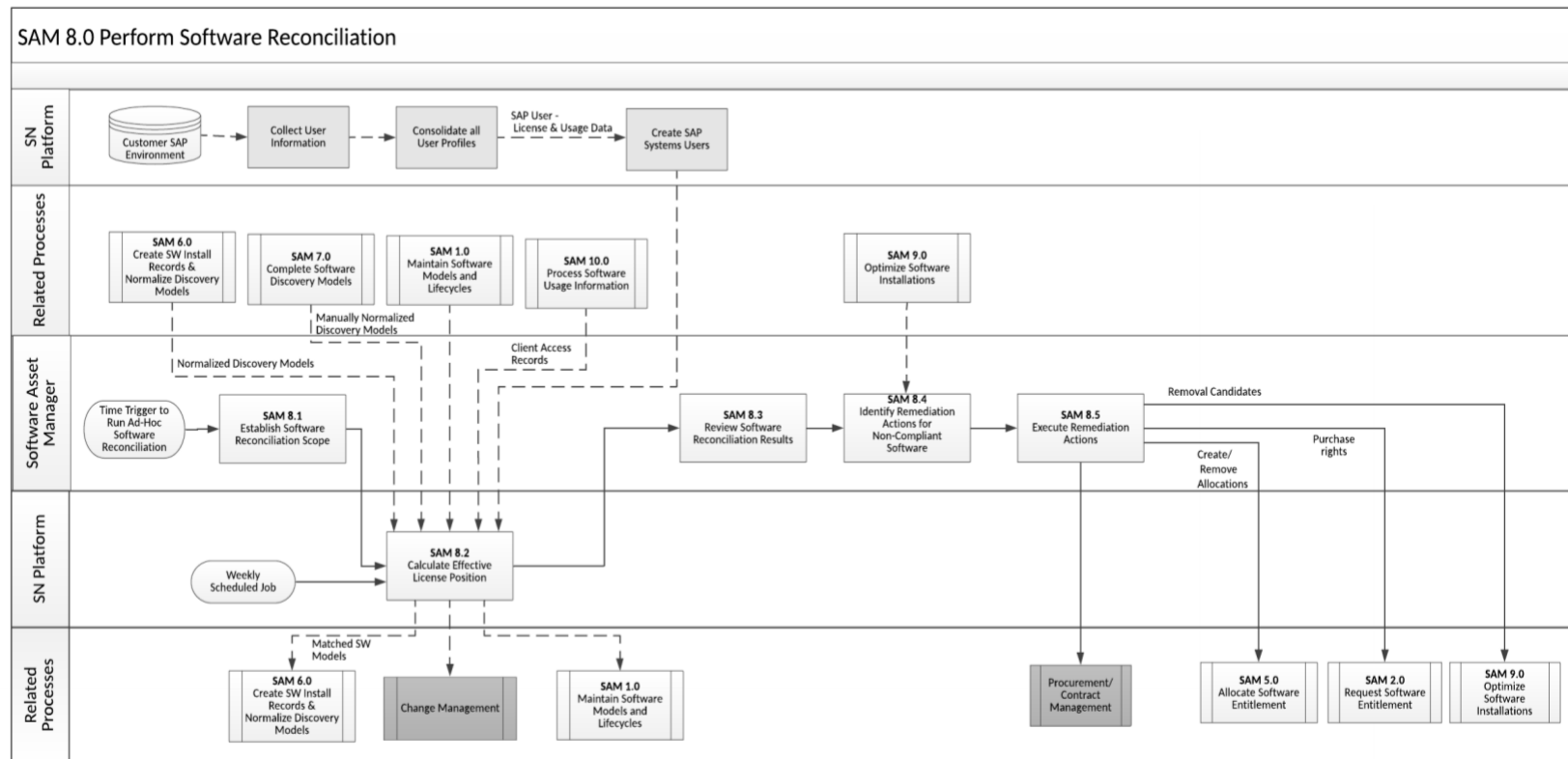
ID	Tasks	Procedures	Primary role	Input	Output
		Link from Create Software Installation Records and Normalize Discovery Models (SAM 6.0) process to review Software Discovery Models following the normalization process.		Rejected software discovery model	
SAM 7.1	Review software discovery models	Review the list of software discovery models. Review as follows: Normalized models – when relevant, identify missing Edition values that require adding to the software installation record. Proceed to SAM 6.3 Partially normalized models and Publisher normalized models – proceed to SAM 7.3 Match Not Found – this occurs when the normalization process cannot fully or partially normalize homegrown or obscure/industry-specific applications not defined in the Software Content Library review the unmatched Discovery Model. Proceed to SAM 7.4.			- Partially normalized discovery model - Software installation edition identified - Unnormalized discovery model
		Link to Create Software Installation Records and Normalize Discovery Models (SAM 6.0) process to add Edition value to an existing software installation record.			Software installation edition identified
		Event trigger to reject normalization rules		Incorrect normalization rules	
SAM 7.2	Reject normalized software discovery model	If, during the execution of SAM processes, it is discovered that the normalization rules assigned to a software discovery model during the automated normalization process (SAM 6.0) are incorrect, de-activate the rules to avoid future usage and re-set/update the Discovery Model for manual update. If the correct	SAM Admin	Incorrect normalization rules	- Software discovery model information - Pattern normalization

ID	Tasks	Procedures	Primary role	Input	Output
		Product/Publisher is not available in the content library, proceed to SAM 7.6 to create pattern normalization rules. Otherwise, proceed to SAM 7.7 to modify the Discovery Model.			rules requirements Custom software product
SAM 7.3	Resolve partially normalized software discovery models	Determine the required information to fully normalize the discovery model (i.e. Product, Version) and modify the model in SAM 7.7. When appropriate, create a custom software product for Publisher Normalized Discovery Models if the correct product is not available in the content library.	SAM Admin	Partially normalized discovery model	- Software discovery model information - Custom software product
SAM 7.4	Resolve unmatched software installations	Determine if the Discovery Model is licensable and whether it requires including for consideration during software reconciliation. If the software requires licensing, create a custom software product record and proceed to SAM 7.6 to create pattern normalization rules that will enable automatic normalization.	SAM Admin	Unnormalized discovery model	Pattern normalization rules requirements
		Link from Maintain Software Models and Lifecycles (SAM 1.0) process because following the creation of a Software Model and discovery mapping conditions associated with a custom product, pattern normalization rules require setting to enable the normalization of the Discovery Model associated with the custom/homegrown software application.		Custom software model record	

ID	Tasks	Procedures	Primary role	Input	Output
SAM 7.5	Create pattern normalization rules	Create and apply pattern normalization rules that apply for custom software products Specify text for Discovery to search for in the software publisher and/or product fields. Specify the normalized attributes (Product, Publisher, Product Type, Platform, Language, Version, etc.) to which the discovered software installation will be normalized when discovered. When applicable create new Publisher and Products if not available in the content library.	SAM Admin	- Custom software model record - Pattern normalization rules requirements	Pattern normalization rules
		Link to Create Software Installation Records and Normalize Discovery Models (SAM 6.0) process to provide pattern normalization rules for custom products into the normalization process	Automated		Pattern normalization rules
		Weekly Time Trigger to review suggested normalization changes based on Central Software Library content.			
SAM 7.6	Review normalization suggestions	Review a Normalization Suggestion to analyze any conflict between what was manually normalized and Central Software Library content. Accept the suggestion to automatically update the Discovery Model to the correct normalized values. Reject the suggestion to keep the manual normalized values.	SAM Admin	Suggested normalization changes	Accepted normalization suggestion Rejected normalization suggestion

ID	Tasks	Procedures	Primary role	Input	Output
SAM 7.7	Modify discovery model	Modify existing Software Discovery Models with the following information: • Product • Version • Platform of Discovered Software (if available) • Language of Discovered Software (if available) • Edition of Discovered Software (if available) • Full Version Number (if available) Once the Discovery Model is updated, the normalization status is set to Manually Normalized. Establish if the manually normalized Discovery Model requires pushing to ServiceNow for inclusion in the Central Software Library. Homegrown applications should be excluded from any data transfer.		• Software discovery model information • Accepted normalization suggestion	Manually normalized discovery model
		Link to Perform Software Reconciliation (SAM 8.0) process to provide manually normalized Discovery Models to match up with entitlements associated with Software Models to calculate effective license positions.	Automated		Manually normalized discovery model
		Daily Scheduled Job to initiate Software Library Content Curation operational process because manually normalized Software Discovery Models require addition to the ServiceNow Central Software Library.			Manually normalized software discovery model

Perform software reconciliation



ID	Tasks	Procedures	Primary role	Input	Output
		Time/Event Trigger to execute and ad-hoc software reconciliation			Ad-hoc software reconciliation requirements
SAM 8.1	Establish software reconciliation scope	Prior to performing and ad-hoc reconciliation, ensure that: • All processor definition updates have been executed in the Central Software Library. • For usage information not automatically derived via platform integrations, ensure all counts related to user and device usage are entered into the appropriate Client Access record. Determine which publishers are in-scope for the ad-hoc software reconciliation. For software reconciliation calculations where compliance is based on the company and/or location of the installation, establish how the reconciliation results require grouping (i.e. Company, Country, Region) Establish if the reconciliation results require grouping by cost center or department to enable chargeback/showback.	SAM Admin	• Ad-hoc software reconciliation requirements • Client access records [Updated] • Processor definitions [Updated]	• Software publisher information • Result grouping dimensions
		Link from Maintain Software Models and Lifecycles (SAM 1.0) process to receive non-SaaS Software Model and linked entitlement records to enable the calculation of effective license positions.	Automated	Software entitlement information	
		Link from Complete Software Discovery Models (SAM 7.0) process to receive manually normalized Discovery Models to match up with entitlements associated with Software Models to calculate effective license positions.	Automated	Software discovery models	
		Link from Create Software Installation Records and Normalize Discovery Models (SAM 6.0) process to receive automatically normalized Discovery Models to match up with entitlements associated with Software Models to calculate effective license positions.	Automated	Software discovery models	
		Link from Process Software Usage Information (SAM 10.0) process to receive software usage information into the calculation of effective license positions of usage-based software licenses.	Automated	Software usage information	

ID	Tasks	Procedures	Primary role	Input	Output
		Link from automated Create SAP Systems Users process to consume System User records (all Named User licenses associated with an individual user) from a SAP ERP environment in order calculate license positions for the SAP Named User license metric by assigning entitlements to the correct System User.	Automated	SAP System user records	
		Link from Monitor SaaS Software Models (SaaSML 1.0) process to receive SaaS Software Model and linked entitlement records to enable the calculation of effective license positions.	Automated	Software entitlement information	
		Weekly Scheduled Job to automatically run software reconciliation for all licensable software. If the reconciliation results require the assignment of grouping dimensions (e.g. group by country/location, etc.), this requires configuring in the automated job. The groupings can be set up in the SAM Pro system properties.	Automated		
SAM 8.2	Calculate effective license position	Based on all automated inputs, calculate effective license positions for selected publishers and produce reconciliation results. This is addressed by matching software installations to the most specific software model based on publisher, product and version. If no match can be found between a software installation and software model, the installation will not be licensed. Rules regarding the rank and order in which compliance is calculated are built into the reconciliation engine to ensure no double counts occur (e.g. an entitlement may be defined as Per User and Per Device). For products identified in the reconciliation with unlicensed installations and no software model exists, a software model record is automatically created so that the unlicensed installations can be associated with a Software Model Notes: - Only 'Licensable' Software Discovery Models are considered during reconciliation. - For software subscription licenses, any use allocations are ignored during reconciliation since the allocations are handled in the vendor's licensing system.	Automated	- Software publisher information - Software entitlement information - Software discovery models - Software usage information - Result grouping dimensions	Software reconciliation results

ID	Tasks	Procedures	Primary role	Input	Output
		- Adobe Creative Cloud software model will be used to reconcile installations of CC and CS versions of Photoshop, Dreamweaver, Illustrator, InDesign, etc. - Only software models that are set as 'License under management' are considered during reconciliation.			
		Link to Maintain Software Models and Lifecycles (SAM 1.0) process to create new Software Models for products discovered with unlicensed installations	Automated		
		Link from Create Software Installation Records and Normalize Discovery Models (SAM 6.0) process because following reconciliation of software products with linked downgrade rights, a specific software model is associated with the software installation record	Automated		Software Model matched with Software Installation Record
SAM 8.3	Review software reconciliation results	Review the software reconciliation results in the License Workbench or in a Publisher Dashboard. Follow attached link for details of the Publisher Analytics Dashboards: https://docs.servicenow.com/bundle/paris-it-asset-management/page/product/software-asset-management2/concept/c_SAMDashSAA.html Reconciliation results are displayed in a layered hierarchy: Product Results - overall compliance status, unlicensed installs, true-up costs, over licensed amounts and removal candidates for a software product. This includes all the publisher's licensable products that have discovered software installs, irrespective of whether entitlements are available. Software Model Results - overall compliance status, unlicensed installs, true-up costs, over licensed amounts removal candidates and remediation options for a software model. License Metric Results – displayed by license metric entitlement (e.g. Per User, Per Device, Per Processor) and displays: Rights owned, Rights used, Unused rights, Over-licensed amount, Allocated in use, Not allocated in use, Allocated not in use, Not allocated, Allocations needed	Software Asset Manager	Software Reconciliation Results	- Reviewed Software Reconciliation Results - Removal Candidates - Compliance Reports

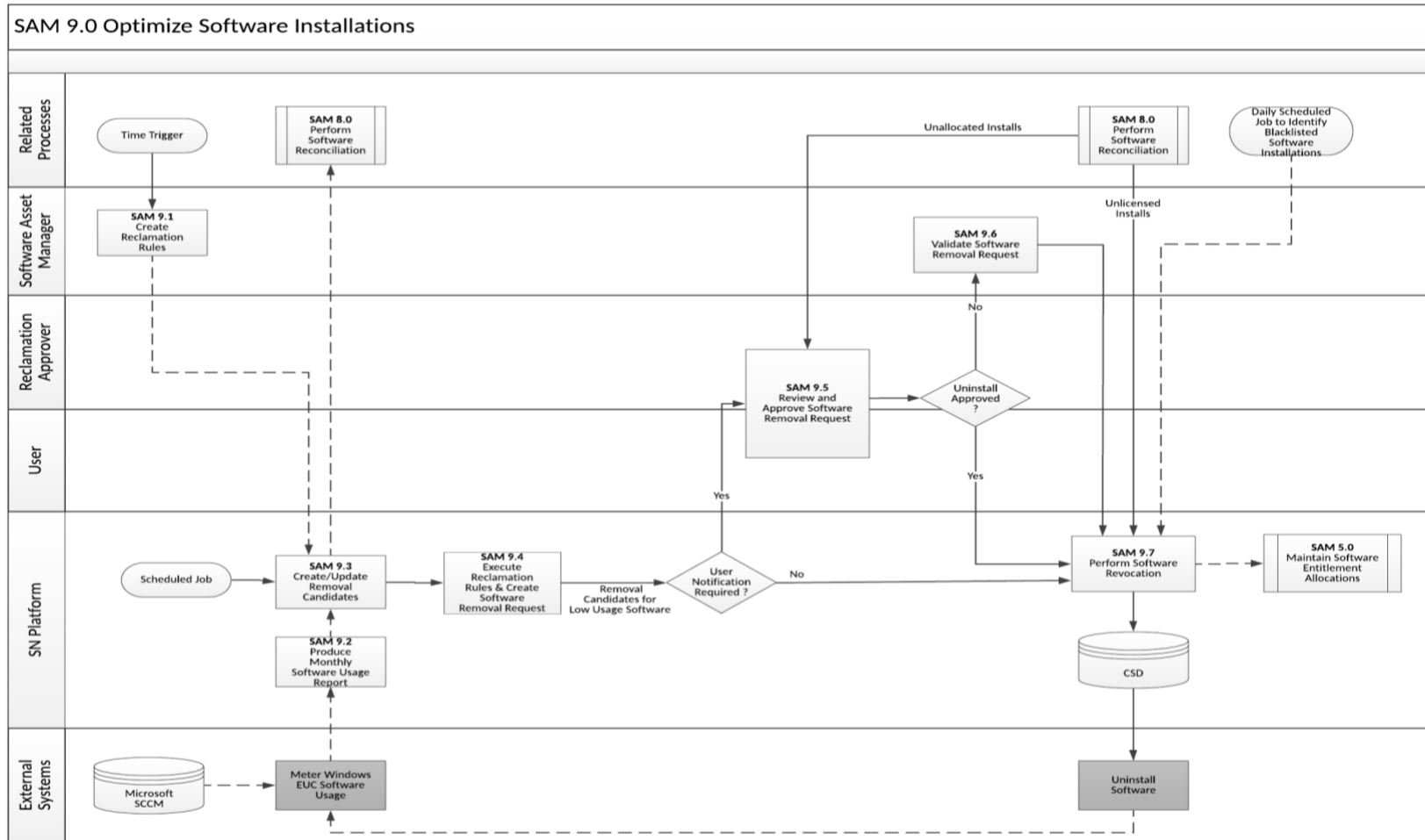
ID	Tasks	Procedures	Primary role	Input	Output
		Rights Used by Record (e.g. details of specific users/devices (Installs Used By) and allocated users/devices (Rights Used By) that consume or have been allocated rights). This result highlights software rights that are being used, where the rights have not been allocated (and vice-versa). The financial values calculated during license compliance are as follows: • Average Price – aggregated cost of entitlements for a specific license metric / number of entitlements purchased • Total Spend - rights owned * average price per right from entitlement per license metric • Potential Savings – number of rights associated with reclamation candidates (unused rights) * average price per right from entitlement per license metric • Over-licensed Amount – rights available * average price per right from entitlement Notes: 1. Common Per Core, Common Per Processor, Microsoft Per Core, Microsoft Per Core with CAL are licensed at the product level. This is to ensure that rights can be shared across software models. All other reconciliation calculations license software installs at the software model level. 2. CAL compliance is associated with the Server Software Model 3. Software models may show as Compliant but include unlicensed installs. This is because the model is part of an Enterprise License Agreement. 4. For Per Named User License Metrics, allocations are created only if the associated CI record has an 'Assigned To' value populated. If no 'Assigned To' values are associated, then the installs are deemed to be non-compliant). 5. For software entitlements associated with the User Subscription license metric, Allocated in use, Not allocated in use, Allocated not in use, Not			

ID	Tasks	Procedures	Primary role	Input	Output
		Allocated, Allocations needed results will always be empty since the subscription software is allocated in the publisher's licensing system/portal. Based on the review, identify non-compliant software positions that require further review. When applicable create reports and circulate them to applicable parties (e.g. SAP Admin, Oracle Admin) to enable user profiles to be updated.			
		Link from Optimize Software Usage (SAM 9.0) process so that removal candidates can be reviewed and actioned during the remediation of non-compliant software (automated).			Removal Candidate Record
SAM 8.4	Identify remediation actions for non-compliant software	Following review of the reconciliation results associated with each non-compliant software product and associated models, analyze the different License Metric results associated with the Software Model result along with the associated remediation options. The following remediation actions are available: • Purchase rights needed for unlicensed software. • Remove unlicensed installs for unlicensed software. • Create allocations for unallocated installs where available rights exist for each unique license metric associated with the Software Model, (except User CAL and Device CAL entitlements) OR create removal candidates for unallocated installs ('not allocated in use'). • Remove allocations for purchased entitlements that have been allocated to a user/device, but the software is not installed ('allocated not in use'). • Claim Removal Candidates identified during the Optimize Software Usage process. Review the remediation actions and determine approaches for creating or removing allocations, creating Removal Candidates for installed software that is unlicensed, reclaiming unused software entitlements for allocation and when appropriate, purchasing new entitlements. Note: Remediation options for User Subscription entitlements do not apply, other than the Purchase rights action.	Software Asset Manager	Reviewed software reconciliation results	Non-compliant software remediation actions

ID	Tasks	Procedures	Primary role	Input	Output
SAM 8.5	Execute remediation actions	For remediating non-complaint software models where available rights exist ('not allocated in use', 'allocated not in use'), perform the following actions: - For 'not allocated in use' (rights purchased but not being used), create allocations from available rights OR create Removal Candidates if it is determined that the unallocated install requires removing from a user/device - For 'allocated not in use', create Removal Candidates if a decision is made to reclaim the rights associated with the allocated install that is not being used. For Removal Candidates identified via software usage monitoring in SAM 9.0, determine if removal is required to remediate a non-complaint software situation and if so, initiate actions to claim the software. For unlicensed software installs (including Device or User CALs), determine the number of rights that require purchasing OR initiate actions to create Removal Candidates to remove unlicensed installations. For over-licensed software (available rights not been used), establish whether the available rights are associated with a maintenance/support contract and if so, engage with procurement/contract management function to determine if the licenses need to be removed from the contract.	Software Asset Manager	Non-compliant software remediation actions	- Software entitlement procurement requirements - Software allocation updates - Claimed software removal candidate - Software removal candidate request
		Link to Allocate Software Entitlement (SAM 5.0) process to create or remove user/device allocations.			Software allocation updates
		Link to Acquire Software Entitlement (SAM 3.0) process to initiate the purchase of entitlement rights to enable software license compliance.			Software entitlement procurement requirements
		Link to Optimize Software Installations (SAM 9.0) process because during the remediation of non-compliant software, it is determined that all software removal candidates require uninstalling and any business rules associated with obtaining user approval are over-ridden.			Claimed software removal candidate

ID	Tasks	Procedures	Primary role	Input	Output
		Link to Optimize Software Installations (SAM 9.0) process because during the remediation of non-compliant software, it is determined that unlicensed or unallocated software installations require uninstalling.			Software removal candidate request
		Link to external Procurement/Contract Management operational process to provide information associated license and maintenance contracts that require renewing and other information that is relevant for the purposes of contract re-negotiation and optimal use of software licenses covered by an enterprise agreement.			Software license contract required

Optimize software installations



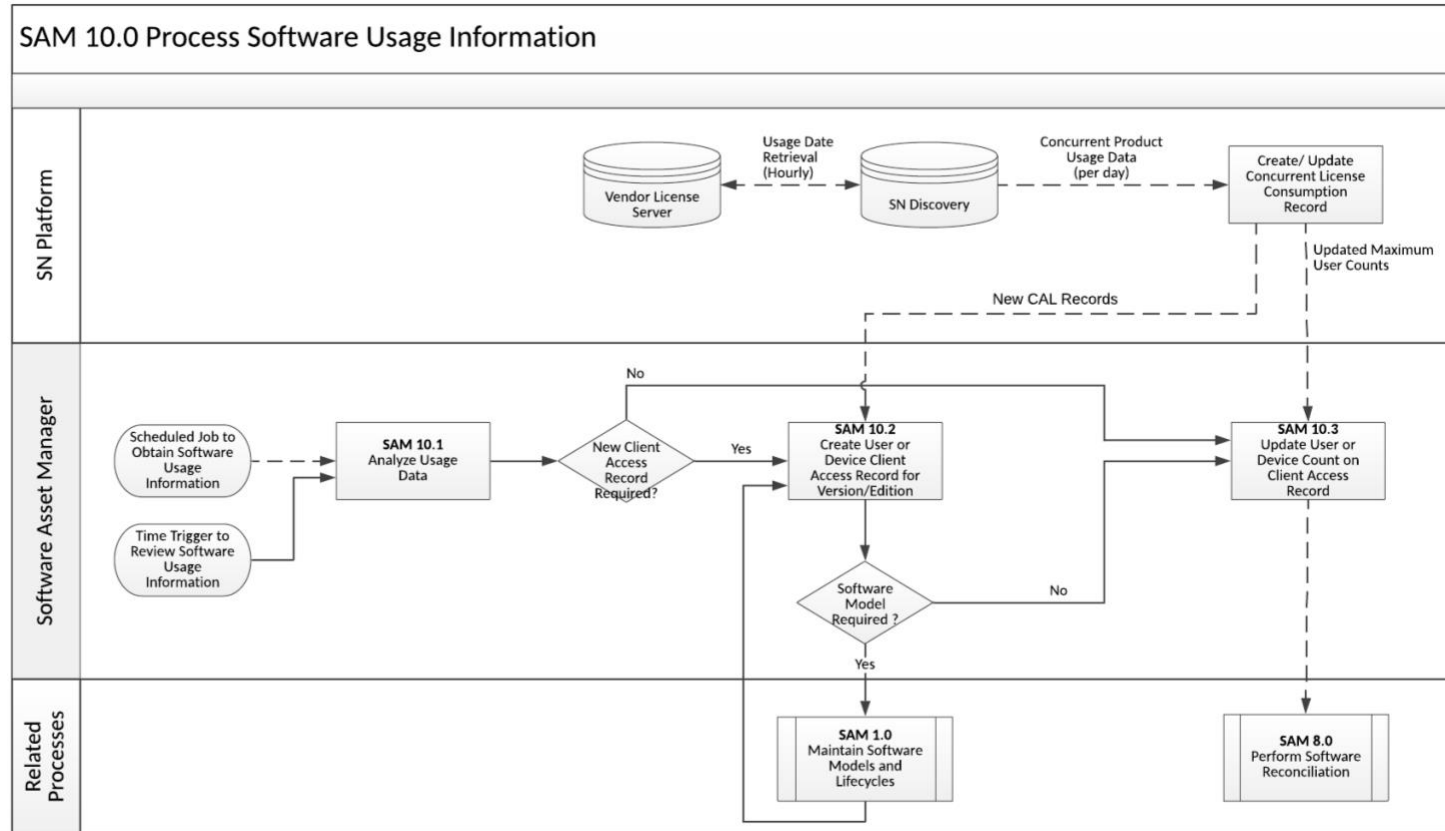
ID	Tasks	Procedures	Primary role	Input	Output
		Time Trigger to create software reclamation rules and specify software usage parameters to provide recommendations for software reclamation or removal.			
SAM 9.1	Create reclamation rules (EUC software only)	Define usage conditions and attributes for software reclamation rules. The rules include: - Name of reclamation rule - Software Products against which the rule needs to be applied - Whether user assigned to the hardware on which the software is installed requires notifying prior to software removal - Number of days after which, if no user response is received, the software is automatically reclaimed - Time period over which to aggregate usage metering information (e.g. last month, last 3 months) and total number of hours the software needs to be used in the defined time period to avoid being reclaimed OR - The last date the software on which must have been accessed to avoid being reclaimed	Software Asset Manager		Reclamation rules
SAM 9.2	Produce monthly software usage report (EUC software only)	Based on software products that are associated with a reclamation rule, produce monthly software usage records based on SCCM usage metering or the date that the software was last used.	Automated with Microsoft SCCM integration	- Software usage information - Reclamation rules	Software usage report
SAM 9.3	Create/update removal candidates	Based on the usage report produced in SAM 9.2 and the usage-related reclamation rules set in SAM 9.1, removal candidates are automatically created via a monthly scheduled job for each software install that violates a reclamation rule or has not been used after	Automated	- Software usage report - Reclamation rules	Removal candidate [New]

ID	Tasks	Procedures	Primary role	Input	Output
	(EUC software only)	a pre-defined date. This triggers the creation of a Software Removal Request (SAM 9.4) For removal candidates where the software has been uninstalled and there are no further software installations that violate the associated reclamation rule, the removal candidate record is closed.			Removal candidate record
		Link to Perform Software Reconciliation (SAM 8.0) process so that removal candidates can be reviewed and actioned during the remediation of non-compliant software.	Automated		Removal candidate record
SAM 9.4	Create software removal request (EUC software only)	Following the creation of a removal candidate, a software removal request is auto created. If the defined reclamation rule is for the software to be uninstalled without notifying the user, the software removal request workflow initiates the software revocation process. Otherwise, request is routed to the user (or user's manager) to obtain approval to reclaim the software.	Automated	Removal candidate	- Software removal request - Approved software removal request
		Link from Perform Software Reconciliation (SAM 8.0) process because during the remediation of non-compliant software, it is determined that unlicensed or unallocated software installations require uninstalling.		Software removal request	
SAM 9.5	Review and approve software removal request	Dependent upon pre-determined policy, the user or business owner of the software may be authorized to approve/reject the removal request OR the user's manager may need to provide a final approval. If the user/business owner or designated approver approves an uninstall OR the specified number of days required for a response (associated with the reclamation rule) is exceeded, this triggers an	Removal Approver	Software removal request	Software removal request [Updated]

ID	Tasks	Procedures	Primary role	Input	Output
		automated action to initiate the revocation of the software installation. If the request is rejected, the removal candidate is reviewed by the Software Asset Manager for a final decision as to whether the software requires uninstalling.			
SAM 9.6	Validate Software removal request	Establish if the software associated with the rejected request should be uninstalled. If it is determined the software requires uninstalling, initiate actions to revoke the software installation. If the Software Asset Manager forces the removal, this triggers an automated action to revoke the software installation.	Software Asset Manager	Software removal request	Approved software removal request
		Link from Perform Software Reconciliation (SAM 8.0) because during the remediation of non-compliant software, it is determined that all unlicensed software installations require uninstalling and any business rules associated with obtaining user approval are overridden (automated).		Claimed software removal candidate	
		Daily Scheduled Job to identify blacklisted software installations which is not authorized for use within an organization. If such installations are identified, a Removal Candidate is created and approved for revocation		Claimed software removal candidate	
SAM 9.7	Perform software revocation	Communicate with Client Software Distribution (CSD) to orchestrate the revocation workflow to uninstall approved or claimed removal and removal candidates from the associated device. Note: If CSD is not configured in the environment, the Removal/Removal Candidate task in a state of Awaiting Revocation can be actioned by technicians to remove the software from the target device.	Automated / Fulfiller	- Claimed software removal candidate - Approved removal candidate	- Software revocation orchestration - Allocation information

ID	Tasks	Procedures	Primary role	Input	Output
		Once the installation is removed from the device, the Removal Candidate is closed (see SAM 9.4).			
		Link to Allocate Software Entitlement (SAM 5.0) process to remove user/device allocations.			Allocation information

Process software usage information

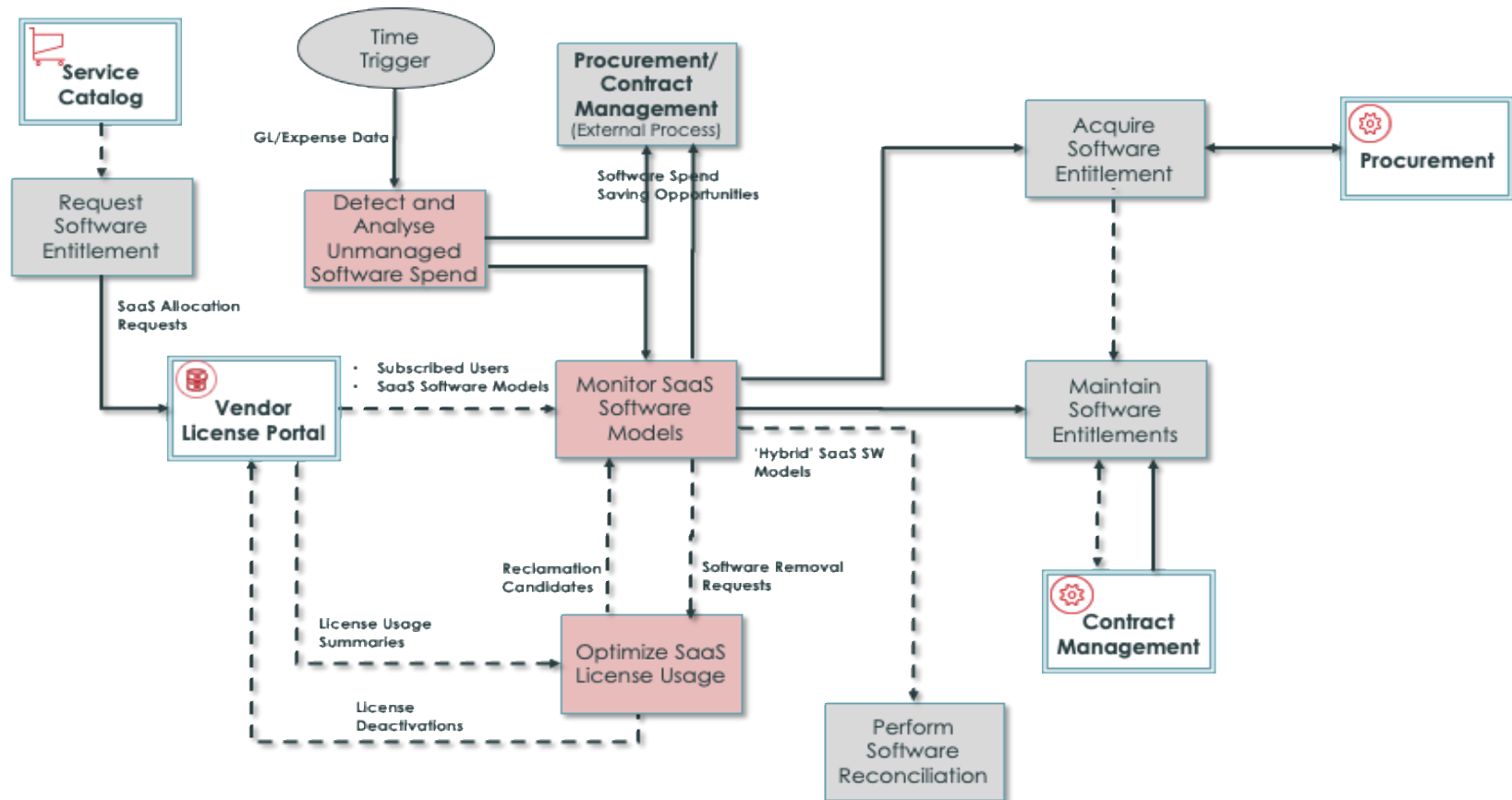


ID	Tasks	Procedures	Primary role	Input	Output
		Scheduled Job to automatically obtain software usage information from systems/portals that are not enabled by OOTB SAM Pro integrations.			Software usage information
		Time Trigger to manually obtain software usage information from a pre-defined source.			Software usage information
SAM 10.1	Analyze usage data	Review software usage information and establish which of the following actions is required: • New client access record required to add user/device counts (requires linking to a server software model) • Update of user/device counts in an existing client access record	SAM Admin	Software usage information	Client access record information
		Link from Maintain Software Models and Lifecycles (SAM 1.0) process following creation of a server software model.		Software model record	
SAM 10.2	Create user or device client access record for version/edition	Following analysis of the received usage information, create the Device or User client access record for the version/edition of the software being used and add the initial user/device counts. This task is automated when SN-Discovery obtains this information from the publisher's licensing server (specifically for Citrix XenDesktop and Xen App concurrent user use cases). If a software model for the server associated with the client access record is not available, initiate actions to create a software model record.	SAM Admin	• Client access record information • Software model record	• Client access record • Software model information
		Link to Maintain Software Models and Lifecycles (SAM 1.0) process because in order to create a client access record, a server software model requires creating.			Software model information
SAM 10.3	Update user or device count on	Based on the usage information received, identify the Device or User client access record and enter the		• Client access record	Client access record [Updated]

ID	Tasks	Procedures	Primary role	Input	Output
	client access record	quantity of devices or users using the server software model. For Citrix XenApp and Xen Desktop use cases, on a daily basis, the maximum number of unique users is obtained from the Citrix licensing server by SN-Discovery and creates/updates a concurrent user consumption record. Each product has single record per day in this table with the maximum in use count for that particular day. if the maximum number of users is greater than the count previously populated, then the client access record is updated with the updated maximum user count. Notes: 1. When creating a User or Device Client Access record for an Oracle database product, specify the Oracle instance that the users or devices have access to. 2. Client access records can also be referenced in the server software model record.		• Software usage information	
		Link to Perform Software Reconciliation (SAM 8.0) process to provide software usage information into the calculation of effective license positions of usage-based software licenses.	Automated		Software usage information

SaaS License Management

SaaS License Management process overview

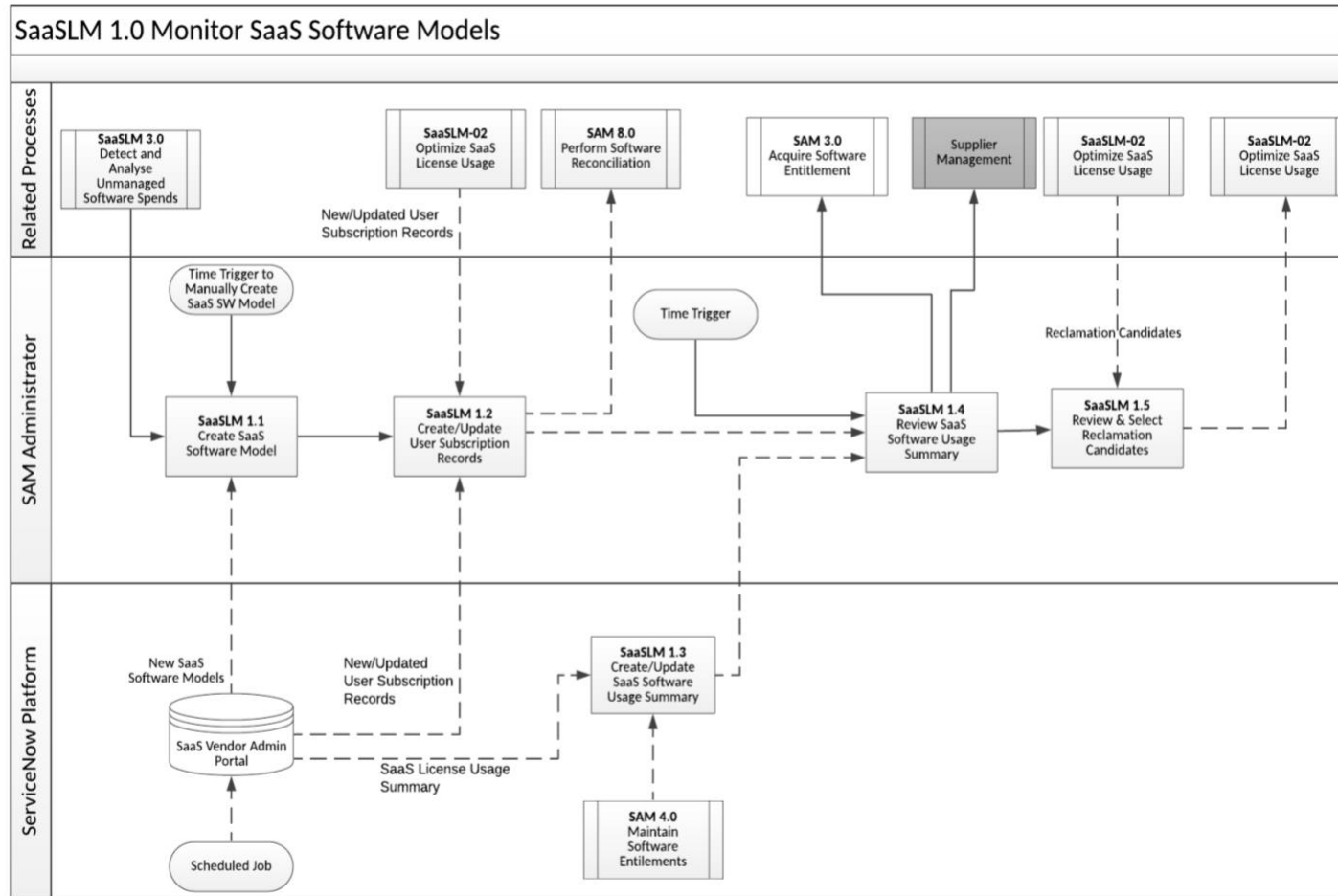


SaaS License Management RACI

ID	Activities	Requester	Approver	Software Asset Manager	SAM Administrator	Procurement Manager	Contract Manager	Finance Admin	Configuration Manager	Software Library Content Curator	Change Manager	Software Catalog Admin
SaaS SLM 1.0	Monitor SaaS Software Models											
SaaS SLM 1.1	Create SaaS Software Model				A/R							
SaaS SLM 1.2	Create/Update User Subscription Records				A/R							
SaaS SLM 1.3	Create/Update SaaS Usage Summary	Automated										
SaaS SLM 1.4	Review SaaS Software Model			C	A/R	C	C					
SaaS SLM 1.5	Review & Identify Reclamation Candidates			R	A/R							
SaaS SLM 2.0	Optimize SaaS License Usage											
SaaS SLM 2.1	Analyse User Activity	Automated										
SaaS SLM 2.2	Assign Reclamation Rules	Automated										
SaaS SLM 2.3	Review/Update Reclamation Rules			A/R	R			C				
SaaS SLM 2.4	Create Reclamation Candidates	Automated										
SaaS SLM 2.5	Review and Approve SaaS License Reclamation	R	A/R	C	I			A/R				
SaaS SLM 2.6	Execute SaaS License Reclamation	Automated										
SaaS SLM 2.7	Transfer User Files	Automated										

ID	Activities	Requester	Approver	Software Asset Manager	SAM Administrator	Procurement Manager	Contract Manager	Finance Admin	Configuration Manager	Software Library Content Curator	Change Manager	Software Catalog Admin
SaaSMLM 2.8	Verify License Deactivation				A/R							
SaaSMLM 2.9	Close Reclamation Candidate	Automated										
SaaSMLM 3.0	Detect & Analyze Unmanaged Software Spend											
SaaSMLM 3.1	Export Data from GL and Expense Systems							A/R				
SaaSMLM 3.2	Load GL and Expense Data into Import Template							A/R				
SaaSMLM 3.3	Discover and Predict Software Spend Transactions									A/R		
SaaSMLM 3.4	Analyze Unnormalized Software Spend Transactions									A/R		
SaaSMLM 3.5	Review Software Spend Detection Results			A/R	C	C	C					
SaaSMLM 3.6	Review Overlapping Software			A/R		C	C					
SaaSMLM 3.7	Resolve Unnormalized Software Spend Transactions			A/R								

Monitor SaaS Software Models



ID	Tasks	Procedures	Primary role	Input	Output
		Event trigger when weekly data imports from a vendor's SaaS portal identify new software models which require creating.	Automated	Software model information	
		Time trigger to manually create new software models for SaaS products not supported with OOTB vendor API integrations.			
		Link to Monitor SaaS Software Models (SaaSML 03) process to create software models for unmanaged products.		Software model information	
SaaSML 1.1	Create SaaS Software Model	This step is automated when an integration exists with a Vendor API to import user subscription data. For SaaS-related Software Models (e.g. Office 365, Adobe CC, DocuSign, Dropbox, Box and Salesforce Cloud), when subscribed products are imported from the vendor's system which are not represented as Software Models, the initial Software Model record is automatically created. For 'hybrid' SaaS products which require the licensing of an actual installation on a device and can be audited for license compliance by publishers (e.g. Office 365, Adobe CC), a Discovery Map is automatically assigned. In situations when an OOTB Vendor API integration is not available, manual creation of a software model record is required. Enter data attributes such as Publisher, Product, Version, Edition, Cost, Product Owner to create Software Model. For hybrid SaaS software models, assign the discovery conditions. Proceed to SaaSML 1.3 to manually add user subscription data.	Automated/ SAM Admin	Software model information	SaaS Software model record [New]
		Weekly Event trigger to import user subscription data via integration with a SaaS vendor's licensing system.		Subscription record	

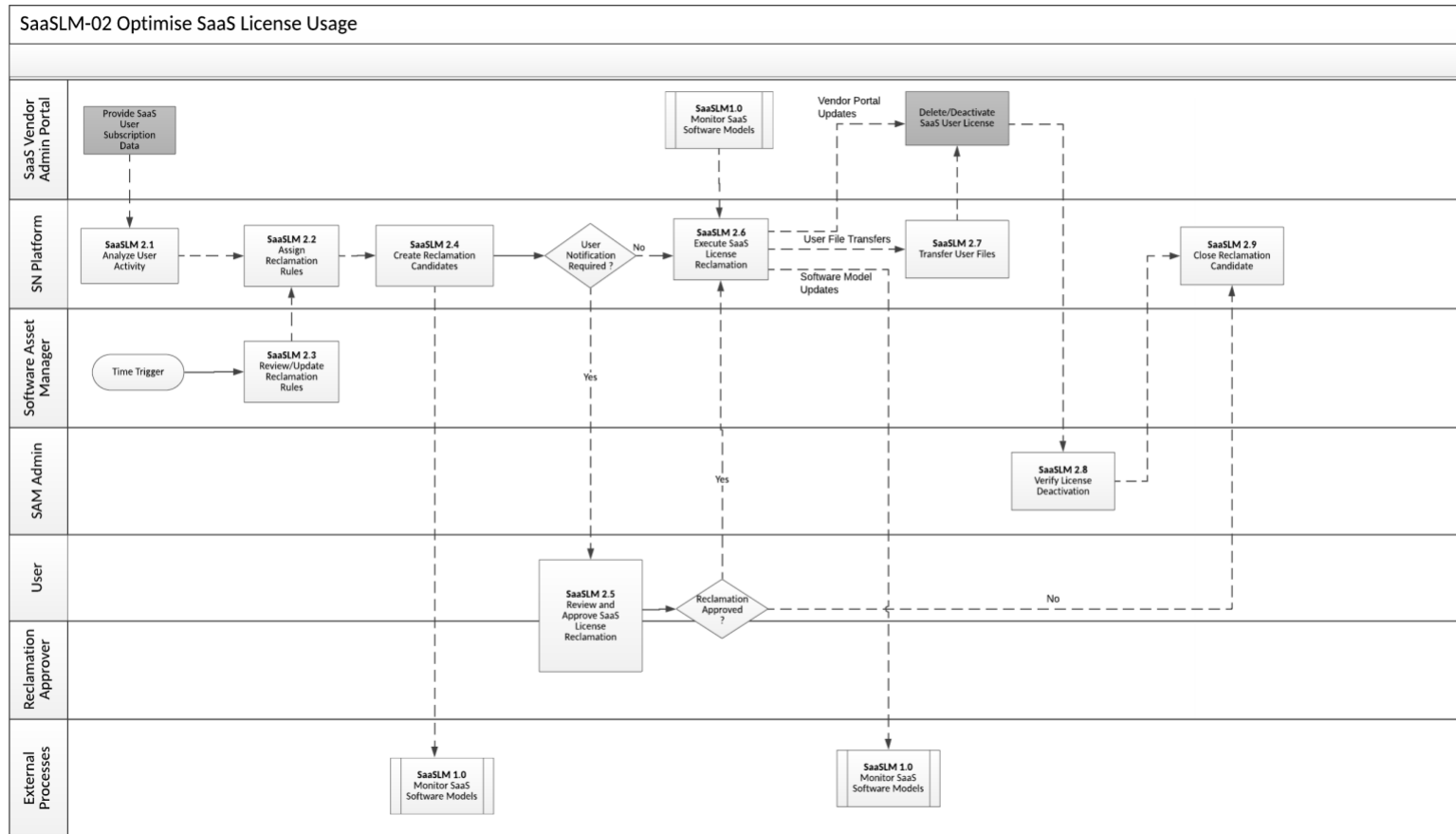
ID	Tasks	Procedures	Primary role	Input	Output
		Link from Optimize SaaS License Usage (SaaS LM 2.0) process to receive updated user subscription information when users are moved to a different license version/edition OR have their subscription deleted.	Automated	Software model updates	
SaaS LM 1.2	Create/Update User Subscription Records	The following information is imported from the SaaS vendor portal and associated with the SaaS software model: - User principal name - Subscription profile - Last activity date If user subscription records require manually adding to the software model, create a software subscription record by entering the following information: - Software model - User Name - Last activity date	Automated/SAM Admin	SaaS Software model record	User Subscription record [New/Updated]
		Link to Perform Software Reconciliation (SAM 8.0) process to calculate compliance for hybrid' SaaS software models/entitlements that are required for licensing discovered installations			- User Subscription record - SaaS Software model record
		Weekly Event trigger to import license usage information via integration with a SaaS vendor's licensing system.	Automated	SaaS License Usage	
		Link from Maintain Software Entitlements (SAM 4.0) process to receive details of the number of rights purchased for population into the license usage summaries.	Automated	Software entitlement record	

ID	Tasks	Procedures	Primary role	Input	Output
SaaS SLM 1.3	Create/update SaaS License Usage Summary	This step is limited to 'pure' SaaS licensing use cases where license compliance is not based on the reconciling of licenses and discovered software installations Usage summaries are recalculated when: • Subscriptions are downloaded from the SaaS vendor's portal • Entitlements change on the software model • Subscriptions are manually created or deleted • Integration profiles are deleted Automatically import the following usage data from SaaS vendor's licensing system: • Total rights owned • Total rights assigned • Total state rights • Total rights available • Average unit cost • Total Spend • Underperforming spend (Average unit cost x state rights – usually actionable at the time of true-up) Notes: • For G-Suite (Docs, Drive and Gmail), the usage summary includes Total suspended rights. This means that licenses has been suspended, but not de-activated. • For DocuSign, enterprise customers license by 'Envelopes' and the DocuSign software model	Automated	SaaS License Usage	Software Model User Summary

ID	Tasks	Procedures	Primary role	Input	Output
		includes an Envelope Consumption tab which provides the following information: • Total number of envelopes • Contract period • Contract start/end date • Units consumed • Units remaining • Expected monthly consumption • Actual monthly consumption			
		Time Trigger to review SaaS software model records to review license usage and reclamation candidates.			
SaaS SLM 1.4	Review SaaS Software Usage Summary	Note: SaaS SLM 1.4 -> SaaS SLM 1.6 are applicable only for 'pure' SaaS software models (e.g. Box, DropBox, DocuSign) that only exist in the cloud where the user subscription is licensed – no device installation is required – and an integration with the SaaS vendor's licensing system provides license usage information (See SaaS SLM 1.3). Review the usage summary and review stale rights and underperforming spend. If it is determined that stale rights require reclaiming, proceed to SaaS SLM 1.5 to review reclamation candidates. https://docs.servicenow.com/bundle/paris-it-asset-management/page/product/software-asset-management2/concept/usage-summary-saas.html If the quantity of rights/envelopes available is below a certain threshold, consumption is greater than expected or a contract is close to expiration, determine if additional rights require procuring. Initiate	Software Asset Manager	Software Model User Summary	• Software entitlement procurement requirements • Software usage reviewed • License renegotiation required

ID	Tasks	Procedures	Primary role	Input	Output
		actions to procure additional SaaS licenses if additional rights are required. If a contract is close to expiration or opportunities are identified to re-negotiate corporate licensing agreements (e.g. downgrading Zoom users from corporate to basic licensing), engage with the appropriate procurement team. Ascertain whether license reclamation can occur prior to any procurement discussions.			
		Link to Acquire Software Entitlement (SAM 3.0) process because additional SaaS software license are required			Software entitlement procurement requirements
		Link to external Supplier Management operational process to provide details of SaaS license usage analysis that have identified opportunities to renegotiate SaaS licensing agreements.			License renegotiation required
		Link from Optimize SaaS License Usage (SaaS LM 2.0) process to analyze low-usage SaaS licenses that are candidates for reclamation	Automated	Reclamation Candidates	
SaaS LM 1.5	Review and Select Reclamation Candidates	Review reclamation candidate record associated with the SaaS software model and select software licenses that can be reclaimed. Take into account the user, the last time the software was used and the 'state' of the reclamation candidate (Awaiting User, Awaiting Approval, Awaiting Revocation) during the process of selecting reclamation candidates. Following review, make final selection of software for reclamation and initiate license deactivation process.	Software Asset Manager	- Reclamation Candidates - Software usage reviewed - User subscription record	License deactivation request
		Link to Optimize SaaS License Usage (SaaS LM 5.0) process to execute SaaS license reclamation			License deactivation request

Optimize SaaS License Usage

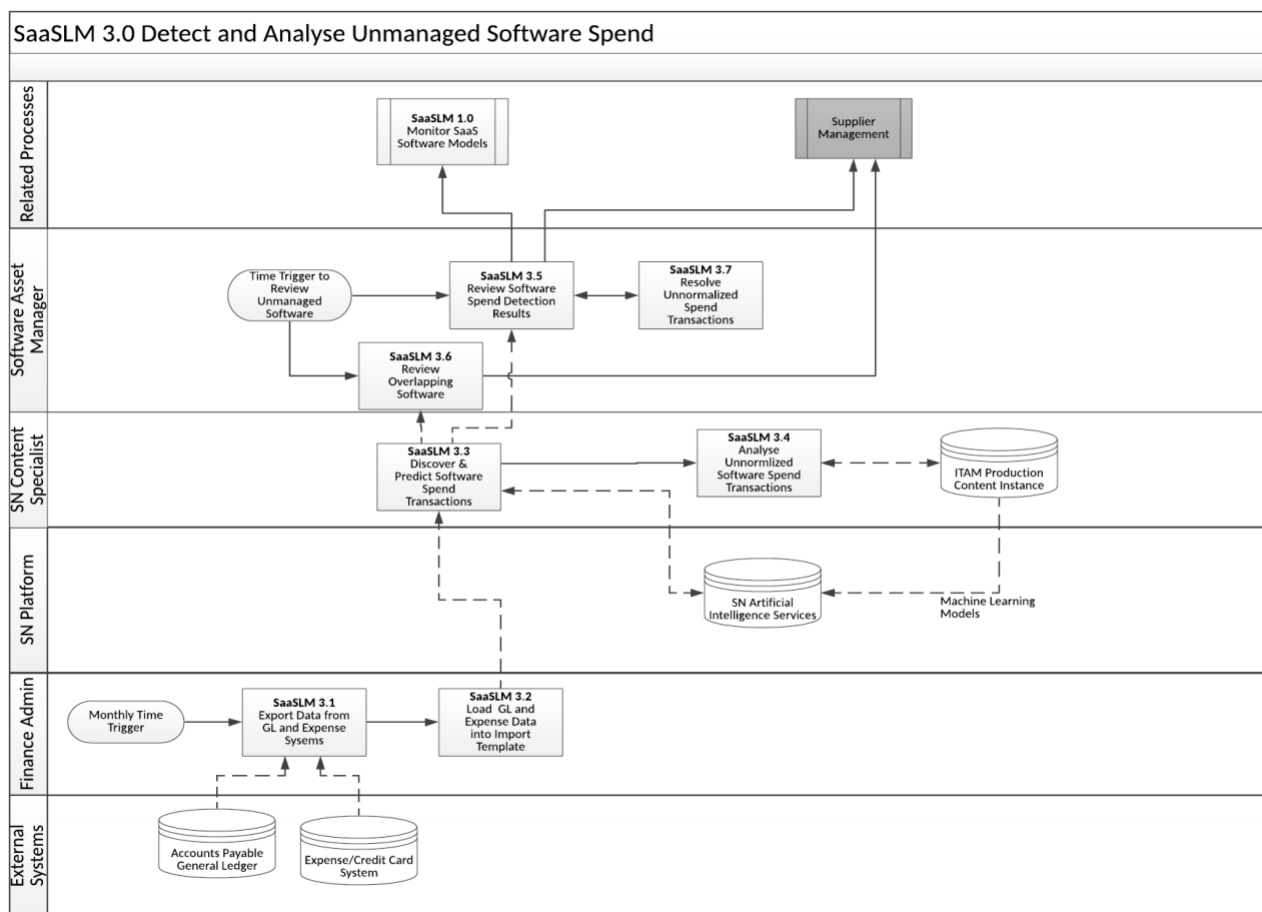


ID	Tasks	Procedures	Primary role	Input	Output
		Link from external SaaS vendor portal to import license usage data for 'pure' SaaS software models	Automated	Software usage information	
SaaSMLM 2.1	Analyse User Activity	Based on user activity detailed in the import, surface stale license information in order to identify reclamation candidates. Note: Following the initial set-up of the integration profiles, historical Salesforce and DocuSign usage data will be immediately synchronized whereas Box and DropBox usage requires analysis of user activity and license staleness information won't get surfaced for 30 days (or the number of days assigned to the reclamation rule) after the integration whilst activity is analyzed in SAM Pro.	Automated	Software usage information	Stale licenses
SaaSMLM 2.2	Assign Reclamation Rules	Assign reclamation rule to stale license.	Automated	- Stale licenses - Reclamation rules	Assigned reclamation rule
		Time trigger to review reclamation rules for SaaS products			
SaaSMLM 2.3	Review/ update reclamation rules	The default rule is 30 days since the last user login date or last meaningful activity performed by the user. Determine if any reclamation rule modifications are required for specific SaaS software products. Other reclamation rule conditions that require confirming: - Whether user assigned to the hardware on which the software is installed requires notifying prior to software removal - Number of days after which, if no user response is received, the software is automatically reclaimed	Software Asset Manager	Reclamation rules	Reclamation rules [Updated]

ID	Tasks	Procedures	Primary role	Input	Output
		In situations when rules require updating, modify the reclamation rule accordingly so that reclamation candidates are created in line with the rules.			
SaaSMLM 2.4	Create reclamation candidate	Based on the usage-based reclamation rules defined in SaaSMLM 2.3 reclamation candidates are automatically created via a monthly scheduled job for each SaaS installation that violates a rule. If the defined reclamation rule is for the software to be uninstalled without notifying the user, the reclamation workflow initiates the license deactivation process. Otherwise, request is routed to the user (or user's manager) to obtain approval to deprovision the SaaS license.	Automated	Assigned reclamation rule	Reclamation candidate
		Link to Monitor SaaS Software Models (SaaSMLM 1.0) process to provide details of low-usage SaaS licenses that are candidates for reclamation	Automated		Reclamation candidate
SaaSMLM 2.5	Review and approve SaaS license reclamation	Dependent upon pre-determined policy, the user of the software may be authorized to approve/reject the removal request OR the user's manager may need to provide a final approval. https://docs.servicenow.com/bundle/paris-it-asset-management/page/product/software-asset-management2/task/reclaim-user-subscription-saas.html If the user or designated approver approves the reclamation OR the specified number of days required for a response (associated with the reclamation rule) is exceeded, this triggers an automated action to initiate the de-activation of the software license.	Automated	Reclamation candidate	• License deactivation request • Rejected reclamation candidate

ID	Tasks	Procedures	Primary role	Input	Output
		Link from Monitor SaaS Software Models (SaaSMLM 1.0) process to execute SaaS license reclamation		License deactivation request	
SaaSMLM 2.6	Execute SaaS License Deactivation	Following the approval of a reclamation candidate or the initiation of the reclamation process via analysis of reclamation candidate in SaaSMLM-01, execute SaaS license deactivation activities which requires for the user subscription to be deactivated or deleted. If the license does not require deactivating in a vendor portal, the reclamation action automatically updates the user subscription records on the software model. For SaaS applications (e.g. G-Suite, Box, Dropbox), files associated with a user profile require moving and preserving before license reclamation occurs. In such situations, proceed to SaaSMLM 2.7. Otherwise, the reclamation workflow will make a call to the API linked to the vendor's licensing admin portal and deprovision/delete the user subscription license OR update the user subscription information on the software model record. Note: When the license reclamation action results in the downgrade of a SaaS license to a lower version/edition (e.g. Zoom use cases), user subscriptions are automatically transferred to the software model associated with the lower version/edition.	Automated	License deactivation request	• User files require transferring • Subscription license requires deleting / deactivating • Software model updates
		Link to external SaaS vendor portal to delete/deactivate user subscription licenses.			Subscription license requires deleting / deactivating
		Link to Monitor SaaS Software Models (SaaSMLM 1.0) process to provide updated user subscription information when users are moved to a different			Software model updates

ID	Tasks	Procedures	Primary role	Input	Output
		license version/edition OR have their subscription deleted.			
SaaSMLM 2.7	Transfer User Files	Transfer user files to a folder prior to account deletion. Specific use cases: Box: User files are transferred to a subfolder of an automatically created folder (ServiceNow – Reclaimed Subscriptions) that resides in the Box admin account. The user is then deleted. Dropbox/DocuSign: User files are transferred to a subfolder of the admin's account. The user is then deleted. GSuite: G-Mail logs require saving/archiving prior to transferring Drive and Docs files and license reclamation.	Automated	User files require transferring	- Subscription license requires deleting / deactivating - User files transferred
		Link to external SaaS vendor portal to delete/deactivate user subscription licenses.		Subscription license requires deleting / deactivating	Deleted/deactivated subscription license
SaaSMLM 2.8	Verify License Deactivation	Verify user subscription license have been deactivated or deleted in the SaaS admin portal.	SAM Admin	Deleted/deactivated subscription license	Deletion/deactivation of subscription license verified
SaaSMLM 2.9	Close Reclamation Candidate	Following the (a) rejection of a request to reclaim a SaaS license OR (b)deactivation/deletion of a user subscription license in the vendor portal, close the reclamation candidate record.	Automated	- Rejected reclamation candidate - Deletion/deactivation of subscription license verified	Closed reclamation candidate



ID	Tasks	Procedures	Primary role	Input	Output
3.0	Detect and Analyse Unmanaged Software Spend	Specific triggers include: • ITAM Gap Analysis (ITAM Driven): Customer wants to see how many software products are managed outside of their purview (e.g. SaaS products that cannot be discovered) and which should be brought in. • Company-Wide Software Inventory (CIO Driven): Companies often want full visibility to see their whole tech stack and will require a software inventory or census to reveal the true footprint of software in the business. • Internal Audit on Software Spending (CFO/Procurement Driven): CFO wants visibility into spending to check for compliance to the company policy. • Reducing Spending on Software (CFO/Procurement Driven): For economic reasons, businesses will demand cost savings on the CIO and overall technology spending, sometimes to avoid layoffs. • Diligence on & integration with acquired company's tech stack: Providing visibility to the tech stack and overlap helps pre & post acquisition.			
SaaSMLM 3.1	Export Data from GL & Expense Systems	Export spend data from GL and/or expense systems to initiate analysis of unmanaged software spend. The volume of spend data is based on what data requires analyzing. This could be the broadest view of spend (all expense and GL transaction) OR a less detailed view (e.g. Software GL category only).	Finance Admin		Exported GL and expense data

ID	Tasks	Procedures	Primary role	Input	Output
SaaSMLM 3.2	Load GL & Expense Data into Import Template	Upload GL and expense data into SAM spend transaction import template with the following spend transactions: • Source • Vendor Name* • Transaction Description* • Expense/GL Account* • Type (Expense or Account payable) * • Cost Center • Department • Location • Amount • Transaction Date • Employee ID *Mandatory information for prediction intelligence function to predict if the transaction is for software along with the publisher and product. Ensure company sensitive data is not included in the import template	Finance Admin	Exported GL and expense data	SAM spend transaction import template [Complete]
SaaSMLM 3.3	Discover and Predict Software Spend Transactions	Using Vendor Name, Transaction Description and GL Account, the SN Predictive Intelligence Engine (shared machine learning capability) discovers and normalizes detected software spend into product categories and predicts publishers/products. If opted into the SAM Content Service, Vendor Name, Transaction Description and GL Account data ('Prediction Data Fields') are shared with ServiceNow content curation function for analysis.	ServiceNow AI Services (Automated)	• SAM spend transaction import template • Software spend normalization rules	• Normalized software spend transactions • Unnormalized software spend transactions • Overlapping software information

ID	Tasks	Procedures	Primary role	Input	Output
SaaSMLM 3.4	Analyse Unnormalized Software Spend Transactions	Review customer's software spend information and identify unnormalized/unlabeled transaction strings (vendor, description, Expense/GL account details) Determine the correct label (publisher, product, software spend = software) for the unlabeled transactions and add rules to the ITAM Production Content instance.	Content Specialist	Unnormalized software spend transactions	Software spend normalization rules
SaaSMLM 3.5	Review Software Spend Detection Results	Review software spend detection results in the Spend Detection Overview UI, review the following information: - Total Detected Spend - Accounts Payable Spend (transactions from corporate systems for software licenses and enterprise/support agreements) - Expense Spend (transactions from an expense system such as Concur) - Unmanaged Publishers (number of publishers associated with unmanaged products) - Unmanaged Products (products that are not represented as software models in SAM Pro) - Unmanaged Spend (total spend associated with products not represented as software models in SAM Pro) Identify any unnormalized detection results and proceed to SaaSMLM-03-07 for further analysis. Analyze expense transactions (i.e. software spend outside of enterprise agreements) across different cost centers, departments and locations and Identify duplicate purchases and whether	Software Asset Manager	- Normalized software spend transactions - Unnormalized software spend transactions - Unnormalized software spend transactions reviewed	- Unnormalized software spend transactions require reviewing - Software spend transactions require addressing - Software model information

ID	Tasks	Procedures	Primary role	Input	Output
		consolidation opportunities exist across departments. Review unmanaged SaaS products and determine if any of them require representing as software models so that they are under SAM control. When appropriate, initiate actions to create a SaaS software model record Provide information of expensed high value/volume software purchases to corporate procurement function to facilitate further action (e.g. cut off one-off subscriptions and consolidate spend with one vendor).			
		Link to Monitor SaaS Software Models (SaaSMLM 01) process to create software models for unmanaged SaaS products.			Software model information
		Link to external Supplier Management operational process to provide information of unmanaged software spend for use in vendor negotiations.			Software spend transactions require addressing
SaaSMLM 3.6	Review Overlapping Software	In the Overlapping Software Overview UI, review overlap of software spend across different product categories (e.g. videoconferencing, CRM). Identify highest spend and highest counts of overlapping software and initiate conversations with procurement/commercial functions (based on accurate data) to enable discussions that will enable the consolidation/elimination of unnecessary software spend.	Software Asset Manager	Overlapping software information	Overlapping software reviewed
		Link to external Supplier Management operational process to provide information of high volume/value software spend so that this			Overlapping software reviewed

ID	Tasks	Procedures	Primary role	Input	Output
		information can contribute to vendor contractual discussions.			
SaaSMLM 3.7	Resolve Unnormalized Software Spend Transactions	Review unnormalized transactions (i.e. publisher product label is not assigned) and when appropriate, assign product, cost center, department and location data attributes.		Unnormalized software spend transactions require reviewing	Unnormalized software spend transactions reviewed

Process control

Process controls are the policies and principles which provide direction over the operation of processes and define constraints or boundaries within which the process must operate.

Name	Description
Policies	Policies and criteria for the inclusion of a component and its attributes in the Software Asset Repository.
Security Policies	Security policies governing access to assets.
Data Exchange Policies	Policy and criteria for the exchange of data between the source hardware asset data and ServiceNow.
Scope	The scope of an organization's Hardware Asset management process is the identification of what is included and what is excluded from this Software Asset Management process.
Change Management Policies	Change Management Policies and procedures.
Management Reports	The frequency and distribution for regularly produced management reports.
Systems Management Architecture	The architecture that provides direction on how Systems Management tools and processes are to be selected, designed and integrated

KPIs

KPIs are best represented as trend lines and tracked over time. They provide information on the effectiveness of the process and the impact of continuous improvement efforts.

KPI/metric	Purpose
% of discovered application installations that have been mapped to normalized application details.	Measure of the effectiveness of the Software Content Library

Number of installations of commercial applications without a license.	Measure of unauthorized installations of non-homegrown software applications
% of application installations which have not been used within the last (say) 90 days.	Identification of removal candidates
Number of installations of applications which are unauthorized (prohibited) for use.	Identification of unauthorized software installed by users that could impact employee productivity and introduce security vulnerabilities to the IT infrastructure.
Value of unlicensed software identified and remediated.	Provides a financial value for unlicensed software remediated by the Software Asset Management team
Number of software vendor audit notifications received per year	Allows for planning of software vendor audits
Costs paid arising from software license audit findings per year	Provides a financial view of the financial consequences of installing software outside of the specified request/procurement process.
Value of software requests fulfilled from using reclaimed software entitlements	Provides details of the financial benefits of a solid software asset management process that evaluates requests for software entitlements prior to any procurement actions.

Appendix A: Glossary of terms and acronyms

Term	Acronym	Definition
Asset		Any resource or capability, including anything that could contribute to the delivery of a service. Assets include hardware, software entitlements and contracts. An asset record contains financial, contractual and lifecycle information.
Attribute		A piece of information about a configuration item. Examples are name, location, version number, and cost. Attributes of CIs are recorded in the configuration management database (CMDB).
Category		A named group of things that have something in common. Categories are used to group similar things together.
Change		The addition, modification, or removal of anything that could have an effect on IT services.
Change Management	CHG	The process responsible for controlling the life cycle of all changes, enabling beneficial changes to be made with minimum disruption to IT services.
Change Record		A record containing the details of a change. Each change record documents the life cycle of a single change. A change record is created for every request for change.
Change Request		See Request for Change.
Configuration Record	CI Record	A record containing the details of a configuration item. Each configuration record documents the lifecycle of a single configuration item. Configuration records are stored in a configuration management database and maintained as part of a configuration management system.
Configuration Item	CI	Any component or other service asset that needs to be managed in order to deliver an IT service. Information about each configuration item is recorded in a configuration record within the configuration management system and is maintained throughout its life cycle by service asset and configuration management. A CI may be: • A physical entity, such as a computer or router • A logical entity, such as an instance of a database • Conceptual, such as a Requisition Service

Configuration Management Database	CMDB	A database used to store configuration records throughout their life cycle. The configuration management system maintains one or more CMDBs. Each CMDB stores attributes of CIs and relationships with other CIs.
Content Service		Customers can opt into the Content Service, anonymously providing software discovery models and processor names that are not normalized, to ServiceNow's internal Content Curation team. This team will then research and update the Software Content Library. When customers choose to opt into the Content Service, they will still have the option to restrict what data is sent to ServiceNow (e.g. home-grown applications will probably be excluded). By having customers share information that is not normalized, we can crowdsource new content and ensure that normalization rates are continuously improved.
Customer		Someone who buys goods or services. The customer of an IT service provider is the person or group who defines and agrees to the service level targets. The term is also sometimes used informally to mean user, for example, 'This is a customer-focused organization.'
Discovery Model		A software discovery model is a 'normalized' version of the discovered software installations that also includes attributes such as Edition, Version and whether the discovered product is licensable.
Effectiveness		A measure of whether the objectives of a process, service, or activity have been achieved. An effective process or activity is one that achieves its agreed objectives. See also Key Performance Indicator.
Efficiency		A measure of whether the right amount of resource has been used to deliver a process, service, or activity.
Key Performance Indicator	KPI	A metric that is used to help manage an IT service, process, plan, project, or other activity. Many metrics may be measured, but only the most important of these are defined as key performance indicators and used to actively manage and report on the process, IT service, or activity. They should be selected to ensure that efficiency, effectiveness, and cost effectiveness are all managed.
License Type		Designates if these are full, outright licenses or upgraded licenses from a previous version or edition (If the Publisher Part Number (PPN) is known, then License type is not required).
License Metric		The calculation by which compliance is determined for the software title.

		The available License metrics are dependent on which Metric groups are available in an instance.
Metric Group		Designates which license metric is used to calculate compliance. A Metric group controls which license metrics are available to be used. It is possible that not all Metric groups will be available in an instance. It is best to check which Metric groups are available, and only select the license metrics available for the Metric group. The Metric group selected does not have to correspond to the publisher of the software title.
Normalization		The normalization process compares the discovered publisher, discovered product, and discovered version values associated with a Discovery Model against the ServiceNow repository of normalized equivalents. Normalized fields are then used to match up entitlements purchased.
Policy		Formally documented management expectations and intentions. Policies are used to direct decisions and to ensure consistent development and implementation of processes, standards, roles, activities, IT infrastructure, and so on.
Publisher Part Number	PPN	The part number of the software title for which rights were purchased. The PPN is the easiest way to ensure success when importing/creating software entitlements. If a part number is not available, the Publisher, Product, Version and Edition fields must be populated. Depending on how the software is purchased, it may be necessary to populate the Platform and Language as well
RACI	RACI	A model used to help define roles and responsibilities. RACI stands for responsible, accountable, consulted, and informed.
Reconciliation		The software reconciliation process calculates the compliance status of software products with respect to discovery and entitlements.
Request for Change	RFC	A formal detailed proposal for a change to be made. The term is often misused to mean change record or the change itself.
Role		A set of responsibilities, activities, and authorities assigned to a person or team. A role is defined in a process or function. One person or team may have multiple roles. For example, a single person may carry out the role of configuration manager and change manager.
Service		A means of delivering value to customers by facilitating the outcomes customers want to achieve without the ownership of specific costs and risks.

Service Level		Measured and reported achievement against one or more service level targets.
Service Level Agreement	SLA	An agreement between an IT service provider and a customer. A service level agreement describes the IT service; documents service level targets and specifies the responsibilities of the IT service provider and the customer.
Service Request		A formal request from a user for something to be provided, for example, a request for information or advice; to reset a password; or to install a workstation for a new user.
Software Asset Management Content Library		The Content Library allows organizations to harness: • Normalization rules to standardize software installations in the ServiceNow® CMDB • Publishers, products, and more to standardize information and eliminate many of the manual activities in the software asset management lifecycle • Pre-built publisher part number definitions and discovery maps to automatically relate purchased software to the correct software installations • Processor definitions to understand correct processor names so that the appropriate processor core factors can be applied during reconciliation
Software Entitlement		Software entitlements ensure that the rights defined in a Software License are being used by the right people/devices, in the right places, at the right time. Software entitlements define license metric details and are assigned to software models. To successfully create a Software Entitlement, Publisher and Product details are required as a minimum.
Software Installation		Discovery tools discover the software <u>installed</u> on a device on an organization's network environment. (e.g. SN Discovery, Microsoft SCCM). Software installation data is often difficult to understand and requires normalization.
Software License		Grants a customer the right to use a particular piece of software. It contains a set of terms and conditions that define to what extent you may legally use that software.
Software Model		Software Models are version specific and represent what an organization has purchased. They are created for all installed software products within an organization and are used to tie software installations (software being used) with entitlements (software

		owned). A Software Model consists of Publisher, Product, Version, Edition, Platform and Language
Software Product		A software title that is version agnostic, e.g. Office, Captivate
Stakeholder		A person who has an interest in an organization, project, or IT service. Stakeholders may be interested in the activities, targets, resources, or deliverables. Stakeholders may include customers, partners, employees, shareholders, owners, or others.
User		A person who uses the IT service on a day-to-day basis. Users are distinct from customers, as some customers do not use the IT service directly.